

FINITE-REFERENCE ERROR BOUNDS FOR RECURRENT EVALUATORS IN POLICY IMPROVEMENT

Anonymous authors

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ABSTRACT

Recurrent value evaluators spend a finite number of internal refinement steps before supporting a policy update. Their computation can remain transient, so evaluation error must be understood without assuming a latent fixed point. We study this problem in Unrolled Policy Iteration with Tiny Recursive Models (UPI-TRM), where a recurrent evaluator guides checker-rewarded plan edits. Our main idea is to compare the deployed depth- n output with a deeper but still finite endpoint at depth m . The deployed value error is bounded by the discrepancy between these endpoints plus the reference endpoint’s self-bootstrap Bellman residual divided by $1 - \gamma^K$. Using standard residual and conservative policy-improvement arguments, we carry this radius through advantage estimation into an exact-mixture improvement bound with signed evaluation error. We also account for propagated target-network lag and approximate mixture realization. For persistent latents, both endpoints are evaluated in one augmented MDP with a shared frozen deployed recurrent map. An exact three-state calculation gives a nonvacuous value certificate and a positive CPI lower bound. The guarantees are conditional on the stated population quantities at one frozen proposal; learned-task performance remains to be established.

1 INTRODUCTION

Tiny Recursive Models (TRMs) refine an internal representation before producing an answer (Jolicoeur-Martineau, 2025). Unrolled Policy Iteration with TRMs (UPI-TRM) uses this inner computation to evaluate plans in a checker-rewarded edit MDP: a value head supplies approximate advantages, a candidate policy chooses edits, and each edit advances the environment. The architectural mapping is in Appendix M (Figure 2).

The evaluator is deployed after n recurrent steps, even when its internal computation is still changing. Comparing its output U_n with a converged recurrence requires a latent fixed point and control of the distance to it. Useful learned computation can instead be transient. Our question is: *can we bound finite-depth evaluation error without assuming recurrent convergence?*

A finite reference. Fix the policy π and compare U_n with the same evaluator at a deeper finite depth $m > n$. The identity

$$U_n - V^\pi = (U_n - U_m) + (U_m - V^\pi)$$

separates sensitivity to additional computation from reference accuracy. Standard Bellman-residual analysis bounds the latter by $\|U_m - \mathcal{T}_K^\pi U_m\|_\infty / (1 - \gamma^K)$, using a K -step current-policy backup that self-bootstraps from U_m . Theorem 5.1 gives the full conditions. A deeper reference helps only when its residual compensates for the endpoint discrepancy; depth alone does not guarantee a tighter bound.

Two depths in one MDP. The Bellman argument treats the endpoints as bounded functions. Recurrence matters when interpreting them as two computation depths of the same deployed evaluator.

With persistent latents, the depth- n map determines what is carried into the successor state. Substituting a depth- m transition would change the MDP. We instead keep the policy and transition kernel fixed and vary only the endpoint used for evaluation.

From evaluation to improvement. The value radius controls advantage error, which enters an exact probability-mixture CPI bound (Kakade & Langford, 2002). Exact centering removes the base-policy component of evaluation error; signed defects retain information lost by absolute worst-case bounds. Target-network lag and approximate deployment are separate corrections.

The contributions are a finite-reference recurrent certificate (Theorem 5.1), its advantage/CPI and target-network connection (Proposition 6.1 and Theorem 7.3), and persistent-state Markovization with a shared frozen map (Corollary C.2). An exact finite-MDP calculation and explicit implementation correspondence complete the analysis. The proof uses established residual and CPI machinery. All guarantees concern one candidate against one frozen base at a fixed snapshot; learned-task performance and training convergence are not established.

2 RELATED WORK

TRMs and HRMs use recurrent latent refinement for structured prediction (Jolicoeur-Martineau, 2025; Wang et al., 2025); deep equilibrium models use an equilibrium representation (Bai et al., 2019). Here the evaluator is read at finite depth, with contraction needed only for the optional fixed-point specialization. Classical approximate dynamic programming controls value error and its propagation through policy or value updates (Bertsekas & Tsitsiklis, 1996; Munos, 2003; Munos & Szepesvári, 2008; Scherrer, 2014). Our primary result instead uses a fixed-policy, population sup-norm residual; Appendix E.1 states the coverage condition for its weighted-norm version.

CPI and safe policy iteration control occupancy shift and approximate improvement (Kakade & Langford, 2002; Kakade, 2003; Pirota et al., 2013; Metelli et al., 2021). We supply the evaluation term for one exact probability mixture from two finite recurrent endpoints. Learning to search casts structured prediction as sequential decisions (Daumé III et al., 2009; Ross et al., 2011; Chang et al., 2015); verifiable-reward systems use automatic correctness signals (DeepSeek-AI, 2025). Our inner depth n is computation within one evaluation, distinct from planning over transitions (Tamar et al., 2016) or the Bellman horizon K . Appendix N.1 preserves the extended comparison, including additional recurrent, control, and constraint-task literature.

3 BACKGROUND AND SETUP

The outer edit MDP. An edit a_t changes plan y_t for a fixed instance x . With h_t edits remaining, the episodic state and nonterminal successor are

$$s_t = (x, y_t, h_t), \quad s_{t+1} = (x, \text{edit}(y_t, a_t; x), h_t - 1).$$

Solving, terminal STOP, or budget exhaustion sends the process to the shared absorber s_{abs} . A bounded checker score $c(x, y)$ supplies reward, optionally shaped as $r = r_0 + \gamma\Phi(s') - \Phi(s)$ with $\Phi = c$ (Ng et al., 1999). We maximize discounted return $\eta(\pi)$ from ρ , with $\gamma \in [0, 1)$. The shaped boundary is $V^\pi(s_{\text{abs}}) = b = -C_{\text{max}}$; folding its tail into the terminal reward with zero terminal bootstrap preserves returns (Appendices G and L).

The inner evaluator. A recurrent map refines a latent while the instance and current plan remain fixed. Starting from $z^{(0)} = z_{\text{init}}(x, y)$ in episodic mode,

$$z^{(k+1)} = \tilde{f}_\theta(z^{(k)}, y, x), \quad U_n(s) = V_\psi(z^{(n)}, x, y).$$

Depth n counts internal computation; the Bellman horizon $K \geq 1$ counts environment transitions. Here $\tilde{f}_\theta = P_{\text{rec}} \circ f_\theta$, where $P_{\text{rec}} = \Pi_R$ is Euclidean projection onto a radius- $R > 0$ ball, or the identity when disabled. Projection can establish an invariant latent set, not strict contraction by itself (Appendix B). The implemented head reads (z, x, y) , not h , but rewards, termination, and successors need the clock.

Reset or carried latent state. Resetting z at each edit makes (x, y, h) Markov. Carrying it requires $\bar{s} = (x, y, z, h)$, since different latents can produce different decisions. The state stores the *pre-unroll* latent; deployment computes $F_n(x, y, z) = \tilde{f}_\theta^{on}(z, y, x)$, uses its output for action selection, and carries it only on nonterminal transitions. Both CPI policies share this frozen map. Figure 1 separates comparison from deployment: $\tilde{U}_m$ starts from stored z without changing F_n , action sampling, or successors.

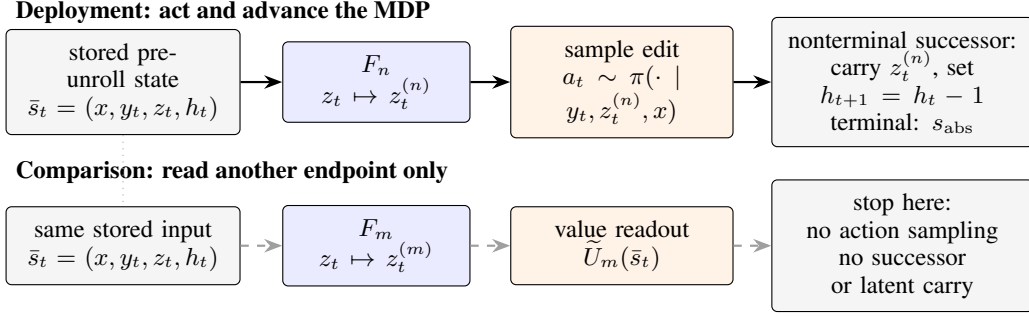


Figure 1: A deeper comparison is not a deeper deployment. Both lanes start from the same stored pre-unroll state. The upper lane uses F_n to choose an action and carries its output only after a nonterminal transition. The lower lane reads another value endpoint and stops: it neither samples an action nor changes the successor. Thus changing m leaves the deployed MDP and trajectory unchanged.

Values and domains. Fix the MDP, recurrence, initialization, value head, and current/candidate policies. State/action spaces are measurable, policies are stochastic kernels, and values are bounded measurable functions. On $\mathcal{C}$, $\|f\|_\infty = \sup_{s \in \mathcal{C}} |f(s)|$ is a pointwise supremum, not a batch maximum or implicit essential supremum. Every endpoint equals b at the absorber; latent paths are defined only elsewhere. Section 5 introduces the advantage-evaluation domain; Appendix E gives all domain definitions.

4 UPI-TRM ALGORITHM

Algorithm 1 freezes the base edit policy and recurrent map before collection. Checker rewards and K -step targets fit the value head; U_n -derived centered advantages fit one candidate policy head. The proposal mixes action probabilities: $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$. Optimizer steps fit this proposal without replacing the base.

The value-derived estimator is

$$\hat{Q}_{U_n}(s, a) = \mathbb{E}[r + \gamma U_n(S') \mid s, a], \quad \hat{A}_{U_n}(s, a) = \hat{Q}_{U_n}(s, a) - \mathbb{E}_{b \sim \pi(\cdot | s)} \hat{Q}_{U_n}(s, b).$$

For persistent latents, condition on the full $\bar{s}$. The theorems analyze the fitted heads at one frozen snapshot. Approximating the mixture by a single network requires the separate deployment correction in Section 8.

Algorithm 1 One fixed-base UPI-TRM proposal on the clock-complete edit MDP

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1: Initialize: sample  $x$ , initialize  $y_0$ , choose  $T \geq 1$ , set remaining budget  $h_0 = T$ , and initialize  $z_0$  for
   persistent- $z$ 
2: for  $t = 0, \dots, T - 1$  do
3:   Set  $s_t = (x, y_t, h_t)$ ; for persistent- $z$ , set  $\bar{s}_t = (x, y_t, z_t, h_t)$ 
4:   Latent input: episodic- $z$  uses  $z_t \leftarrow z_{\text{init}}(x, y_t)$ ; persistent- $z$  uses the carried pre-unroll  $z_t$ 
5:   Evaluator:  $z_t^{(n)} \leftarrow \tilde{f}_\theta^{\text{on}}(z_t, y_t, x)$ 
6:   Value:  $U_n(s_t) \leftarrow V_\psi(z_t^{(n)}, x, y_t)$  in episodic- $z$ ;  $\tilde{U}_n(\bar{s}_t) \leftarrow V_\psi(z_t^{(n)}, x, y_t)$  in persistent- $z$ 
7:   Action: sample  $a_t \sim \pi_\phi(\cdot \mid y_t, z_t^{(n)}, x)$  and apply the edit; observe  $r_t, y_{t+1}$ , and terminal flag  $d_t$ ; set
    $h_{t+1} = h_t - 1$ 
8:   if  $d_t$  or  $h_{t+1} = 0$  then
9:     Store  $(s_t, a_t, r_t, s_{\text{abs}})$  in episodic- $z$ , or  $(\bar{s}_t, a_t, r_t, s_{\text{abs}})$  in persistent- $z$ 
10:    break
11:   else
12:     Set  $s_{t+1} = (x, y_{t+1}, h_{t+1})$  and  $\bar{s}_{t+1} = (x, y_{t+1}, z_t^{(n)}, h_{t+1})$ 
13:     Store  $(s_t, a_t, r_t, s_{t+1})$  in episodic- $z$ , or  $(\bar{s}_t, a_t, r_t, \bar{s}_{t+1})$  in persistent- $z$ 
14:     For persistent- $z$ , carry the post-unroll latent:  $z_{t+1} \leftarrow z_t^{(n)}$ 
15:   end if
16: end for
17: Value fit: form  $G^{(K)}$  and regress the mode-appropriate evaluator ( $U_n$  or  $\tilde{U}_n$ )
18: Advantage estimate: compute  $\hat{A}$  centered conditional on  $s_t$ , or on full  $\bar{s}_t$  in persistent- $z$ 
19: Candidate fit: take policy-gradient steps to obtain one  $\pi_{\text{cand}}$  while the base policy and recurrent map
   remain frozen
20: Mixture construction:  $\pi_\alpha \leftarrow (1 - \alpha)\pi_\phi + \alpha\pi_{\text{cand}}$ 
21: Optional deployment heuristic: distill  $\pi_\alpha$  into a single policy network

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Only nonterminal transitions carry a latent; every terminal cause ends collection after its final record. Appendix L gives folded targets and replay details. Exact baseline enumeration ensures statewise centering; approximate integration instead contributes a signed defect (Table 6, Appendix J). One depth- n unroll per action term over T states costs $O(T|\mathcal{A}|n)$ recurrent-map applications. Interpolating weights, logits, or hidden states, or distilling the mixture, generally changes its law.

5 FINITE-REFERENCE EVALUATION AND ADVANTAGE ERROR

We first use episodic notation; Section 8 applies the same argument on the fixed augmented MDP.

5.1 THE FINITE-REFERENCE VALUE CERTIFICATE

The triangle inequality splits $\|U_n - V^\pi\|_\infty$ into endpoint discrepancy and reference error. Bellman contraction controls the latter through its self-bootstrap residual (Puterman, 1994; Sutton & Barto, 2018).

For later advantage estimation, let $\mathcal{C}_\pi$ carry the initial law and be invariant under π . Write $\mathcal{C} = \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ for a measurable set containing $\mathcal{C}_\pi$, the absorber, and successors of one current or candidate action, followed by any current-policy continuation (Definition E.2). Define $\mathcal{C}^\circ = \mathcal{C} \setminus \{s_{\text{abs}}\}$ and $\|f\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} = \sup_{s \in \mathcal{C}} |f(s)|$. On this invariant domain the residual bound is

$$\|V - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{1}{1 - \gamma^K} \|V - \mathcal{T}_K^\pi V\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}, \quad (1)$$

Apply this estimate to $V = U_m$ and add the endpoint discrepancy:

Theorem 5.1 (Measurable finite-reference-depth value decomposition). *Let $\mathcal{C}$ be a measurable state set with its trace σ -algebra, let P_π be a Markov kernel satisfying $P_\pi(\mathcal{C} \mid s) = 1$ for every $s \in \mathcal{C}$, and fix $\gamma \in [0, 1)$ and an integer $K \geq 1$. If an initial law is used, assume $\rho(\mathcal{C}) = 1$. Assume the one-step reward is uniformly bounded and its fixed-policy conditional mean $r^\pi(s) := \mathbb{E}_s^\pi[r_0]$ is measurable on $\mathcal{C}$. Then the K -step expected reward*

$$r_K^\pi(s) := \mathbb{E}_s^\pi \left[\sum_{i=0}^{K-1} \gamma^i r_i \right] \quad (2)$$

is bounded and measurable on $\mathcal{C}$. On the Banach space $B_b(\mathcal{C})$ of bounded measurable functions with the ordinary pointwise sup norm, define

$$(\mathcal{T}_K^\pi V)(s) := r_K^\pi(s) + \gamma^K (P_\pi^K V)(s). \quad (3)$$

Then $\mathcal{T}_K^\pi : B_b(\mathcal{C}) \rightarrow B_b(\mathcal{C})$ is a γ^K -contraction and has a unique fixed point V^π . For arbitrary endpoint functions $U_n, U_m \in B_b(\mathcal{C})$ and finite integers $0 \leq n < m$,

$$\|U_n - V^\pi\|_\infty \leq \underbrace{\|U_n - U_m\|_\infty}_{\text{finite-depth discrepancy}} + \underbrace{\frac{\|U_m - \mathcal{T}_K^\pi U_m\|_\infty}{1 - \gamma^K}}_{\text{reference-depth residual}}. \quad (4)$$

For the recurrent specialization, take $\mathcal{C} = \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ and $\mathcal{C}^\circ = \mathcal{C} \setminus \{s_{\text{abs}}\}$. On $\mathcal{C}^\circ$, fix a jointly measurable recurrent map $(s, z) \mapsto T(s, z)$, a common measurable initialization into its declared latent domain, and a latent norm, and define $z^{(j)}$ by iterating that map and $U_j(s) = V_\psi(z^{(j)}(s), x(s), y(s))$. At the absorber define only the common scalar boundary $U_j(s_{\text{abs}}) = b$, and assume the two endpoint functions lie in $B_b(\mathcal{C})$. If the measurable scalar head is L_V -Lipschitz in its latent argument uniformly while (x, y) is held fixed, then

$$\|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq L_V \sup_{s \in \mathcal{C}^\circ} \|z^{(n)}(s) - z^{(m)}(s)\| \quad (5)$$

$$\leq L_V \sup_{s \in \mathcal{C}^\circ} \sum_{j=n}^{m-1} \|z^{(j+1)}(s) - z^{(j)}(s)\| \quad (6)$$

$$\leq L_V \sum_{j=n}^{m-1} \sup_{s \in \mathcal{C}^\circ} \|z^{(j+1)}(s) - z^{(j)}(s)\|. \quad (7)$$

If $\mathcal{C}^\circ$ is empty, the nonnegative latent suprema in this chain are defined to be zero.

The discrepancy measures sensitivity to additional computation; the residual measures agreement with the frozen policy's value equation. A stable output can still have a large residual. Both endpoints remain finite, and the backup uses the same policy and kernel as V^π . The Lipschitz clause converts latent separation to value separation. Taking the supremum after the path sum is sharper than summing separate suprema. Proof in Appendix K.

Appendix N.2 gives the noncontractive examples, additional endpoint bounds, and frozen-schedule extension.

Uniform recurrent contraction gives the optional geometric L_z^n limit in Appendix B. An alternative weighted- L_p certificate appears in Appendix E.1; it requires explicit domination of the fixed-policy occupancy by the reference measure. Neither specialization is needed for the finite-endpoint result.

5.2 FROM VALUE ERROR TO CANDIDATE ADVANTAGE ERROR

The action value and baseline each contribute at most γE from successor value error E , giving $2\gamma E$. Policy overlap sharpens the candidate-specific bound. Define

$$\tau(s) := \text{TV}(\pi_{\text{cand}}(\cdot | s), \pi(\cdot | s)), \quad \tau_\infty := \sup_{s \in \mathcal{C}_\pi} \tau(s) \leq 1. \quad (8)$$

We use $b(s) = \mathbb{E}_{\pi_{\text{cand}}}[\hat{A}(s, a) - A^\pi(s, a)]$ for the signed candidate defect, and $\varepsilon_{A, \text{cand}} = \sup_{s \in \mathcal{C}_\pi} |b(s)|$. The function $b(s)$ is distinct from the scalar absorbing boundary b . The normalized discounted occupancy is $d_\pi = (1 - \gamma) \sum_{t \geq 0} \gamma^t \rho P_\pi^t$.

Value error is controlled on $\mathcal{C}$, advantage error on $\mathcal{C}_\pi$. The check is post-candidate because $\mathcal{C}$ contains candidate successors.

Corollary 5.2 (Advantage-error bridge). *Under the hypotheses of Theorem 5.1, define*

$$B_{n,m} := \|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{\|U_m - \mathcal{T}_K^\pi U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}{1 - \gamma^K}. \quad (9)$$

Let $\hat{A} = \hat{A}_{U_n}$ be the exact one-step centered estimator of Eq. (145), with the transition expectation evaluated exactly (or applied to a deterministic transition) and the baseline computed exactly under π . Then Assumption 7.1 holds with the valid choice

$$\varepsilon_A = 2\gamma B_{n,m}. \quad (10)$$

If Assumption B.1 holds on $\mathcal{D}^\circ = \mathcal{C}^\circ$ and the L_V -Lipschitz head condition also holds there, sending $m \rightarrow \infty$ gives the fixed-point specialization

$$\varepsilon_A \leq \underbrace{\frac{2\gamma \varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{architectural}} + \underbrace{2\gamma L_V \frac{L_z^n}{1 - L_z}}_{\text{unrolling}} C_z. \quad (11)$$

The reference assesses the depth- n estimator without changing its action selection. Exact centering gives the sharper candidate-average bound:

Corollary 5.3 (Candidate evaluation bias from policy overlap). *Let $\hat{A}_V$ be the exactly centered one-step estimator, $e = V - V^\pi$, and $\delta_V = \|e\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Use τ and τ_∞ from Eq. (8). Then, for every $s \in \mathcal{C}_\pi$,*

$$\left| \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [\hat{A}_V(s, a) - A^\pi(s, a)] \right| \leq 2\gamma \delta_V \tau(s), \quad (12)$$

and hence

$$\varepsilon_{A, \text{cand}} \leq 2\gamma \delta_V \tau_\infty, \quad (13)$$

$$\mathbb{E}_{s \sim d_\pi} |b(s)| \leq 2\gamma \delta_V \mathbb{E}_{s \sim d_\pi} \tau(s). \quad (14)$$

For an exactly centered K -step estimator whose intervening actions follow π , replace γ by γ^K . The fallback $2\gamma \delta_V$ follows from $\tau \leq 1$.

Proof in Appendix N.3.

Exact centering cancels the shared part of the two action laws. Consequently, $2\gamma \delta_V \tau(s)$ can be much smaller than the worst-case $2\gamma \delta_V$, and is zero when the policies agree. Section 7 uses this candidate bias in an improvement bound. In particular, under the exact-mixture and centering hypotheses of Theorem 7.3, the finite-reference radius gives

$$\eta(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma\tau_\infty}{1 - \gamma} B_{n,m} - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (15)$$

Here $\hat{L}_\pi$ is the estimated CPI surrogate and ε_{CPI} bounds the magnitude of the candidate’s expected true advantage on the mixture domain; both are defined in Section 7. This display completes the link from finite recurrent endpoints to policy improvement. For centered K -step estimators the advantage factor becomes $2\gamma^K$; other estimators, including GAE, need their own error bound (Remark J.3).

6 TARGET-NETWORK POPULATION-RESIDUAL BRIDGE

Value regression fits U_n to a target network $\bar{V}$, whereas the certificate needs a backup of the evaluated function itself. For a frozen current-policy trajectory, the target is $G^{(K)} = \sum_{k < K} \gamma^k r_k + \gamma^K \bar{V}(S_K)$, with loss $\mathcal{L}_{\text{val}} = \mathbb{E}_{S_0 \sim \mu} [(U_n(S_0) - \text{sg}(G^{(K)}))^2]$. Here μ selects starting states; the conditional trajectory law is π . Even zero target residual can leave a self-bootstrap residual when the networks differ. The correction is propagated target lag:

Proposition 6.1 (Target-network population residual bridge). *Let $\mathcal{E}$ be a measurable P_π -invariant state domain. Assume $r_K^\pi \in B_b(\mathcal{E})$ and $U_q, \bar{V} \in B_b(\mathcal{E})$ for a finite depth $q \geq 0$, and suppose the conditional K -step target trajectory follows the frozen policy π . Define*

$$\varepsilon_{\text{targ}, q} := \|U_q - (r_K^\pi + \gamma^K P_\pi^K \bar{V})\|_{\infty, \mathcal{E}}. \quad (16)$$

Retain the propagated target lag by defining

$$R_{\text{targ}, q}^P := \varepsilon_{\text{targ}, q} + \gamma^K \|P_\pi^K (\bar{V} - U_q)\|_{\infty, \mathcal{E}}, \quad (17)$$

$$R_{\text{targ}, q}^{\text{sup}} := \varepsilon_{\text{targ}, q} + \gamma^K \|\bar{V} - U_q\|_{\infty, \mathcal{E}}. \quad (18)$$

Then the self-bootstrap residual satisfies

$$\|U_q - \mathcal{T}_K^\pi U_q\|_{\infty, \mathcal{E}} \leq R_{\text{target}, q}^P \leq R_{\text{target}, q}^{\text{sup}}. \quad (19)$$

For the formal absorbing representation, require $U_q(s_{\text{abs}}) = \bar{V}(s_{\text{abs}}) = b$ and use the same absorbing reward and boundary in both population operators. The equivalent folded implementation may instead fold γb into the terminal-transition reward and mask the terminal bootstrap to zero, but it must not add the absorbing tail again.

Proof in Appendix N.4.

Propagated lag averages before taking the supremum; the ordinary sup-lag is a possibly looser fall-back. Both vanish when the functions coincide. Table 7 (Appendix N.6) and Appendix I.2 give the value/CPI substitutions. These are population sup norms: even population mean-square regression controls only an $L^2(\mu)$ conditional residual. Empirical loss or a finite-batch maximum establishes neither the uniform residual nor uniform propagated lag.

7 CONSERVATIVE POLICY IMPROVEMENT

The exact mixture $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$ can select its base or candidate component with a Bernoulli draw at each decision. CPI corrects the base-policy surrogate for advantage error and occupancy shift (Kakade & Langford, 2002); we keep signed error and bound shift using this selector. Error uses $\mathcal{C}_\pi$; occupancy needs $\mathcal{C}_{\text{mix}}$ to contain full trajectories under both policies, not just one deviation (Appendix E, Table 4). The uniform-error assumption below suffices for the scalar specialization but is not needed for the signed bound.

Assumption 7.1 (Uniform advantage error on $\mathcal{C}_\pi$). For some $\varepsilon_A \geq 0$:

$$|\hat{A}(s, a) - A^\pi(s, a)| \leq \varepsilon_A \quad (20)$$

for every $s \in \mathcal{C}_\pi$ and for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every a .

Write $d_\pi = (1 - \gamma) \sum_{t \geq 0} \gamma^t \rho P_\pi^t$ and define the true and estimated CPI surrogates

$$\begin{aligned} L_\pi(\pi') &:= \eta(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi'(\cdot | s)} [A^\pi(s, a)], \\ \hat{L}_\pi(\pi') &:= \eta(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi'(\cdot | s)} [\hat{A}(s, a)]. \end{aligned}$$

Lemma 7.2 (Exact-mixture occupancy). *Let π and π_{cand} act in the same MDP from the same initial law, and let π_α be their exact pointwise probability mixture. For $\alpha \in [0, 1]$,*

$$\text{TV}(d_{\pi_\alpha}, d_\pi) \leq \frac{\gamma\alpha}{1 - \gamma + \gamma\alpha}. \quad (21)$$

For finite or countable state spaces, with $\text{TV}(p, q) = \frac{1}{2} \|p - q\|_1$,

$$\|d_{\pi_\alpha} - d_\pi\|_1 \leq \frac{2\gamma\alpha}{1 - \gamma + \gamma\alpha}. \quad (22)$$

Proof in Appendix N.5.

The order- α occupancy shift multiplies the candidate's order- α surrogate contribution. The resulting correction is $O(\alpha^2)$ for a fixed domain and advantage span. The evaluation correction combines base-policy centering defect $c(s)$ and candidate bias $b(s)$ under d_π , the surrogate's occupancy.

Theorem 7.3 (Occupancy-aware CPI with signed evaluation error). *Fix one discounted MDP with bounded rewards and measurable current and candidate policy kernels. Let $\alpha \in [0, 1]$ and let $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$ be their exact pointwise probability-space mixture, with no parameter interpolation, hidden-state interpolation, logit interpolation, or distillation. Let $\mathcal{C}_{\text{mix}}$ be measurable with $\rho(\mathcal{C}_{\text{mix}}) = 1$ and invariant under both π and π_α , as in Definition E.3. Assume that a measurable π -invariant domain $\mathcal{C}_\pi \subseteq \mathcal{C}_{\text{mix}}$ also carries ρ . Assume the functions below are bounded and*

measurable on their declared domains. Define g on $\mathcal{C}_{\text{mix}}$ and c, b on $\mathcal{C}_\pi$ by

$$g(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)}[A^\pi(s, a)], \quad \Delta_g := \sup_{s \in \mathcal{C}_{\text{mix}}} g(s) - \inf_{s \in \mathcal{C}_{\text{mix}}} g(s), \quad (23)$$

$$c(s) := \mathbb{E}_{a \sim \pi(\cdot|s)}[\hat{A}(s, a)], \quad b(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)}[\hat{A}(s, a) - A^\pi(s, a)], \quad (24)$$

$$\Xi_\alpha := \mathbb{E}_{s \sim d_\pi}[(1 - \alpha)c(s) + \alpha b(s)]. \quad (25)$$

Then the surrogate error is exactly

$$\hat{L}_\pi(\pi_\alpha) - L_\pi(\pi_\alpha) = \frac{\Xi_\alpha}{1 - \gamma}, \quad (26)$$

and

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{\Xi_\alpha}{1 - \gamma} - \frac{\gamma\alpha^2\Delta_g}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (27)$$

Because Ξ_α is signed, a usable one-sided guarantee may replace it by any established $B_\alpha \geq \Xi_\alpha$, for example

$$B_\alpha = (1 - \alpha)\mathbb{E}_{d_\pi}|c(s)| + \alpha\mathbb{E}_{d_\pi}|b(s)|. \quad (28)$$

Under exact statewise centering,

$$\mathbb{E}_{a \sim \pi(\cdot|s)}[\hat{A}(s, a)] = 0 \quad \text{for every } s \in \mathcal{C}_\pi, \quad (29)$$

define the uniform candidate-policy bias

$$\varepsilon_{A, \text{cand}} \triangleq \sup_{s \in \mathcal{C}_\pi} \left| \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)}[\hat{A}(s, a) - A^\pi(s, a)] \right|. \quad (30)$$

If $\varepsilon_{A, \text{cand}} < \infty$ and $\varepsilon_{\text{CPI}} := \sup_{s \in \mathcal{C}_{\text{mix}}} |g(s)| < \infty$, then

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{\alpha}{1 - \gamma} \varepsilon_{A, \text{cand}} - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}, \quad (31)$$

Exact centering leaves candidate bias scaled by $\alpha/(1 - \gamma)$. Occupancy shift uses Δ_g and the exact denominator $1 - \gamma + \gamma\alpha$. A negative defect credits underestimated advantage; an absolute bound loses that credit (Section 9). Any established $B_\alpha \geq \Xi_\alpha$ gives a one-sided guarantee. Sampled/learned baselines contribute $c(s)$, even with a small batch mean. Proof in Appendix I.

The domain and span may depend on α . A common-domain safe-step extension appears in Corollary N.3 (Appendix N.7); Appendix I.1 gives its proof and boundaries. Assumption 7.1 implies $\varepsilon_{A, \text{cand}} \leq \varepsilon_A$, but the signed theorem does not require it.

8 PERSISTENT STATE AND DEPLOYMENT REALIZATION

Markovization with a carried latent. On $\bar{s} = (x, y, z, h)$, F_n determines both policies' post-unroll inputs and the nonterminal successor $(x, \text{edit}(y, a; x), F_n(x, y, z), h - 1)$. Termination reaches the common absorber without carrying a latent. Keeping this frozen map while evaluating $\tilde{U}_m$ from the same stored z gives bounded endpoints in one MDP, so Theorem 5.1 yields

$$\|\tilde{U}_n - V^{\bar{\pi}}\|_\infty \leq \|\tilde{U}_n - \tilde{U}_m\|_\infty + \frac{\|\tilde{U}_m - \tilde{T}_K^{\bar{\pi}}\tilde{U}_m\|_\infty}{1 - \gamma^K}. \quad (32)$$

The norm includes the common scalar boundary on the augmented advantage-evaluation domain. Corollary C.2 gives the full conditions and target/CPI forms. Slow drift is required only for the optional memoryless fixed-point comparison.

Approximating the mixture at deployment. Distillation or parameter/logit/hidden-state interpolation generally gives a different law $\tilde{\pi}$. On a domain invariant under both policies, let its uniform TV discrepancy from π_α be at most δ_{dep} , and let Δ_r be the common reward-interval width. Under the standard-Borel and measurable-kernel assumptions of Theorem H.5,

$$|\eta(\tilde{\pi}) - \eta(\pi_\alpha)| \leq \frac{\Delta_r \delta_{\text{dep}}}{(1 - \gamma)(1 - \gamma + \gamma\delta_{\text{dep}})}. \quad (33)$$

The first-mismatch penalty vanishes for exact realization. Either-direction uniform KL control supplies a TV radius (Corollary H.7). Persistent policies must share the recurrent transition map; changing it requires a different-MDP analysis.

9 EXACT FINITE-MDP NONVACUITY CHECK

Exact enumeration gives a nonvacuous value radius and signed CPI bound, but a vacuous absolute specialization. Nothing is sampled or learned.

Consider two nonabsorbing states and a shared absorber, $\mathcal{S} = \{s_0, s_1, s_{\text{abs}}\}$, actions $\{a, b\}$, initial law $\rho = \delta_{s_0}$, $\gamma = 1/2$, and $K = 1$. The current policy always chooses a and the candidate always chooses b . Deterministic reward-successor pairs are

$$(s_0, a) : (0, s_0), \quad (s_0, b) : (1, s_1), \quad (s_1, a) : (0, s_0), \quad (s_1, b) : (0, s_1),$$

and both absorber actions return $(0, s_{\text{abs}})$. The complete domains are $\mathcal{C}_\pi = \{s_0, s_{\text{abs}}\}$ and $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} = \mathcal{C}_{\text{mix}} = \mathcal{C}_{\text{pair}} = \mathcal{D}_{\text{dep}} = \mathcal{S}$. For the scalar recurrence, set $z^{(0)} = 0$, $T_s(z) = u_s - z/2$, $(u_{s_0}, u_{s_1}) = (1/4, -1/4)$, and $u_{s_{\text{abs}}} = 0$, with $V_\psi(z, s) = z$. Use deployed depth $n = 1$, comparison depth $m = 2$, and $\alpha = 1/2$. Table 1 reports the exact result.

Table 1: Exact two-nonabsorbing-state calculation. State vectors are ordered $(s_0, s_1, s_{\text{abs}})$. No quantity is sampled or learned.

Quantity	Exact value
$V^\pi; U_1; U_2$	$(0, 0, 0); (1/4, -1/4, 0); (1/8, -1/8, 0)$
$\ U_1 - U_2\ _\infty; \ U_2 - \mathcal{T}_1^\pi U_2\ _\infty$	$1/8; 3/16$
Finite-reference upper bound; actual $\ U_1 - V^\pi\ _\infty$	$1/2; 1/4$
$d_\pi; d_{\pi_{1/2}}; \pi_{1/2}(a s), \pi_{1/2}(b s)$	$(1, 0, 0); (3/4, 1/4, 0); (1/2, 1/2)$ at every state
$g; \Delta_g; c; b$	$(1, 0, 0); 1; (0, 0, 0); (-1/4, -1/4, 0)$
$\hat{L}_\pi(\pi_{1/2}); \Xi_{1/2}$	$3/4; -1/8$
Signed CPI lower bound; actual $\eta(\pi_{1/2})$	$2/3; 3/4$

The finite-reference radius $B_{1,2} = 1/2$ is twice the actual error $1/4$; it uses two finite endpoints rather than a recurrent limit.

The policy calculation illustrates why signs and spans matter. The estimator understates candidate advantage: $b(s_0) = -1/4$ and $\Xi_{1/2} = -1/8$. Correcting the surrogate $3/4$ therefore *adds* $1/4$. The exact-span occupancy penalty is $1/3$, so

$$\eta(\pi_{1/2}) \geq \frac{3}{4} + \frac{1}{4} - \frac{1}{3} = \frac{2}{3}, \quad \eta(\pi_{1/2}) = \frac{3}{4}.$$

The absolute uniform specialization instead subtracts $1/4$ for evaluation error and replaces the span $\Delta_g = 1$ by $2\varepsilon_{\text{CPI}} = 2$, doubling the occupancy penalty:

$$\frac{3}{4} - \frac{1}{4} - \frac{2}{3} = -\frac{1}{6}.$$

Both bounds are valid, but only the signed-span version certifies improvement over $\eta(\pi) = 0$. This is an exact nonvacuity check, not a learned-model performance result. Appendix F derives every quantity; `verify_finite_mdp.py` reproduces the rational calculation.

10 DISCUSSION, LIMITATIONS, AND CONCLUSION

A finite reference separates endpoint discrepancy from self-bootstrap residual and supplies a value radius for one-proposal CPI. Target lag and deployment mismatch require separate corrections; persistent state requires the clock and a shared frozen map. Recurrent convergence is unnecessary.

These conditional guarantees establish neither optimizer/policy-sequence convergence, finite-sample coverage, near-optimality, nor learned-task improvement. The exact MDP checks nonvacuity. Certification still requires uniform residual, centering/defect, and deployment controls; finite-population diagnostics do not establish them. Appendix N.8 gives the extended interpretation and proposed certification program. Ethics, reproducibility, and AI-use statements follow.

AI USE STATEMENT

Generative AI tools assisted with literature search, revision of the manuscript’s exposition, interpretation of the exact finite-MDP example, figure editing, LaTeX reorganization, and preparation of formatting, cross-reference, and exact-arithmetic validation checks. This statement records assistance in manuscript preparation; responsibility for the submission remains with the authors.

REPRODUCIBILITY STATEMENT

The assumptions are stated with each result, and complete proofs appear in Appendices I–K and N, as well as alongside the appendix extensions. Appendix F gives the full rational calculation for Table 1; the standard-library script `verify_finite_mdp.py` reproduces its quantities. Appendix L specifies the implementation semantics.

ETHICS AND BROADER IMPACT

Ethics statement. The paper studies a mathematical model of checker-trained Tiny Recursive Models on synthetic plan-editing tasks. We use no personal data, no human subjects, no crowdsourcing, and no deployed decision system. The main ethical risk is rhetorical: conditional bounds can be overstated as evidence about general reasoning or high-stakes automation. We report no empirical capability claim and state explicitly that the bounds do not certify improvement for a learned model.

Broader impact. The methodological contribution is to separate internal evaluator depth from outer policy improvement, which may help analyses of verifier-feedback systems with bounded internal compute. We make no deployment recommendation and keep the claims conditional on the displayed assumptions.

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A SUPPLEMENTARY MATERIAL

Appendices B and C develop optional recurrent limits and the persistent-state construction. Appendices D–G collect notation, domains, the exact example, and terminal accounting. Appendix H uses the edit clock to obtain finite-horizon and deployment bounds. The remaining appendices provide proofs and implementation details.

Assumption A.1 (Bounded rewards). Potential-shaped rewards are uniformly bounded. For some $R_{\max} < \infty$,

$$|r(s, a, s')| \leq R_{\max} \quad \text{for every transition.}$$

For $R > 0$, define the Euclidean projection

$$\begin{aligned} \Pi_R(z) &= \arg \min_{u: \|u\|_2 \leq R} \|u - z\|_2 \\ &= \begin{cases} z, & \|z\|_2 \leq R, \\ R \frac{z}{\|z\|_2}, & \|z\|_2 > R. \end{cases} \end{aligned} \quad (34)$$

$$G^{(K)}(s_0) = \sum_{k=0}^{K-1} \gamma^k r_k + \gamma^K \bar{V}(s_K). \quad (35)$$

$$C_z := \sup_{s \in \mathcal{C}^\circ} \|\tilde{f}_\theta(z_{\text{init}}(x(s), y(s)), y(s), x(s)) - z_{\text{init}}(x(s), y(s))\| < \infty. \quad (36)$$

B OPTIONAL RECURRENT-CONTRACTION SPECIALIZATIONS

This appendix adds structure to the finite-reference result. Uniform contraction on an invariant latent set gives a recurrent fixed point and turns the finite path into a geometric tail. The primary theorem needs neither this limit nor the following contraction assumption.

Projection onto B_R is nonexpansive and establishes invariance when the initializer lies in that ball. Strict contraction additionally requires a proved modulus below one for the complete projected map. Operator-norm clamping or local finite differences may guide an intervention, but their use alone is not such a proof. Projection-disabled mode uses the identity; setting $R = 0$ would instead collapse every latent to zero.

Assumption B.1 (Domain-relative forward-invariant contraction). Fix a declared nonabsorbing state domain $\mathcal{D}^\circ$. The latent space $\mathcal{Z}$ is a complete normed space (in our implementation, $\mathbb{R}^d$ with the Euclidean norm). For $s \in \mathcal{D}^\circ$, set

$$T_s(z) := \tilde{f}_\theta(z, y(s), x(s)).$$

There exist a closed set $\mathcal{Z}_{\text{inv}} \subseteq \mathcal{Z}$ and $L_z \in [0, 1)$ such that, uniformly for every $s \in \mathcal{D}^\circ$,

$$T_s(\mathcal{Z}_{\text{inv}}) \subseteq \mathcal{Z}_{\text{inv}}, \quad \|T_s(z) - T_s(z')\| \leq L_z \|z - z'\| \quad (z, z' \in \mathcal{Z}_{\text{inv}}).$$

Every initialization used by the result satisfies $z_{\text{init}}(x(s), y(s)) \in \mathcal{Z}_{\text{inv}}$ for every $s \in \mathcal{D}^\circ$.

For the episodic finite-depth and projection-aware specializations, invoke Assumption B.1 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and assume the finite C_z in Eq. (36). The persistent slow-drift specialization declares its augmented domain in Appendix C.2. A global contraction and initialization condition over every input pair is sufficient, but it is not implied by a closure-restricted premise. We use $L_z^0 = 1$ for every $L_z \in [0, 1)$ and $L_z^n = 0$ for every integer $n \geq 1$ when $L_z = 0$.

Proposition B.2 (Domain-relative fixed point and finite-unrolling bias (Banach)). *Under Assumption B.1, for every $s \in \mathcal{D}^\circ$ the map T_s has a unique fixed point $z^*(s) \in \mathcal{Z}_{\text{inv}}$. For every $z^{(0)}(s) \in \mathcal{Z}_{\text{inv}}$, its iterates converge linearly:*

$$\|z^{(t)}(s) - z^*(s)\| \leq L_z^t \|z^{(0)}(s) - z^*(s)\|. \quad (37)$$

If $\mathcal{D}^\circ = \mathcal{C}^\circ$, C_z is finite as in Eq. (36), and $z^{(0)}(s) = z_{\text{init}}(x(s), y(s))$, then uniformly on $\mathcal{C}^\circ$,

$$\|z^{(n)}(s) - z^*(s)\| \leq \frac{L_z^n}{1 - L_z} C_z. \quad (38)$$

Thus fixed-point existence is asserted only for input pairs represented by the declared domain. Assumption B.1's metric contraction and invariance clauses alone do not imply measurability of $s \mapsto z^*(s)$.

For a sufficient measurability condition, equip the closed set $\mathcal{Z}_{\text{inv}}$ with its Borel σ -algebra and $\mathcal{D}^\circ$ with its declared σ -algebra. If

$$T : \mathcal{D}^\circ \times \mathcal{Z}_{\text{inv}} \longrightarrow \mathcal{Z}_{\text{inv}}, \quad T(s, z) = T_s(z), \quad (39)$$

is jointly measurable and $z^{(0)} : \mathcal{D}^\circ \rightarrow \mathcal{Z}_{\text{inv}}$ is measurable, then every iterate $z^{(j)}$ and the fixed-point map z^* are measurable.

For any result below that invokes L_V on a declared nonabsorbing domain $\mathcal{D}^\circ$, let $\mathcal{Z}_{\mathcal{D}}(s)$ be the set of latent arguments compared by that result at s . Assume the scalar head is uniformly Lipschitz while the instance and plan are held fixed: for every $s \in \mathcal{D}^\circ$ and all $z, z' \in \mathcal{Z}_{\mathcal{D}}(s)$,

$$|V_\psi(z, x(s), y(s)) - V_\psi(z', x(s), y(s))| \leq L_V \|z - z'\|. \quad (40)$$

For a contractive specialization one may take $\mathcal{Z}_{\mathcal{D}}(s) = \mathcal{Z}_{\text{inv}}$; a pathwise result needs the condition only at the latent pairs it displays.

Assumption B.3 (Domain-explicit bounded value heads). For an invocation on a declared nonabsorbing state domain $\mathcal{D}^\circ$, declare a latent set $\mathcal{Z}_{\text{bd}} \subseteq \mathcal{Z}_{\text{inv}}$. It is either all of $\mathcal{Z}_{\text{inv}}$ or an explicitly identified smaller set containing every latent used by that result. There exists $V_{\text{max}} < \infty$ such that, for every represented $s \in \mathcal{D}^\circ$ and every $z \in \mathcal{Z}_{\text{bd}}$,

$$|V_\psi(z, x(s), y(s))| \leq V_{\text{max}}, \quad |V_{\bar{\psi}}(z, x(s), y(s))| \leq V_{\text{max}}. \quad (41)$$

The absorber is handled by its separately declared finite scalar boundary; no absorbing latent is introduced. This assumption can supply boundedness in the optional contractive specializations, but it does not replace any explicit $B_b(\mathcal{C})$ premise in a Bellman theorem.

For the optional contractive specialization, invoke Assumption B.1 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and the finite constant C_z above. For each represented $s \in \mathcal{C}^\circ$, write $z^*(s)$ for the unique fixed point of T_s . The joint measurability and measurable initialization make $s \mapsto z^*(s)$ measurable by Proposition B.2. Invoke Assumption B.3 on $\mathcal{D}^\circ = \mathcal{C}^\circ$ with $\mathcal{Z}_{\text{bd}} = \mathcal{Z}_{\text{inv}}$; together with the separate finite absorbing boundary, this makes U_* bounded on the closure. Set $U_*(s_{\text{abs}}) = -C_{\text{max}}$ and define

$$\varepsilon_{\text{res}}^* \triangleq \|U_* - \mathcal{T}_K^\pi U_*\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (42)$$

Consecutive latent differences satisfy $\|z^{(j+1)} - z^{(j)}\| \leq L_z^j C_z$ uniformly. Hence Theorem 5.1 gives

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq L_V C_z \frac{L_z^n (1 - L_z^{m-n})}{1 - L_z} + \frac{\|U_m - \mathcal{T}_K^\pi U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}{1 - \gamma^K}. \quad (43)$$

Uniform convergence $U_m \rightarrow U_*$ and Bellman contraction imply convergence of the residual norm. Taking $m \rightarrow \infty$ recovers

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \underbrace{\frac{\varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{fixed-point residual}} + L_V \underbrace{\frac{L_z^n}{1 - L_z} C_z}_{\text{geometric path bound}}. \quad (44)$$

This limit is a specialization, not an assertion that arbitrary TRMs are designed to converge. Under $\|z_{\text{init}}\|_2 \leq R$ with $\mathcal{Z}_{\text{inv}} = B_R$, projection gives $C_z \leq 2R$ (Remark K.2).

When every represented pre-projection output satisfies $\|f_\theta(z, y(s), x(s))\|_2 \geq \rho_R > R$ for $z \in B_R$, radial projection reduces the pre-projection Lipschitz upper bound by R/ρ_R . The projected map is contractive only when $(R/\rho_R)L_z^{\text{pre}}(R) < 1$. Proposition K.3 gives

$$L_z \leq \bar{L}_z^{\text{sat}}(R) := \frac{R}{\rho_R} L_z^{\text{pre}}(R), \quad (45)$$

and Proposition K.4 gives, for $B_0(x, y) := V_\psi(0, x, y)$ and $B_0(s_{\text{abs}}) := -C_{\text{max}}$,

$$\varepsilon_{\text{res}}^* \leq \varepsilon_0 + (1 + \gamma^K) L_V(R) R, \quad (46)$$

where $\varepsilon_0 := \|B_0 - \mathcal{T}_K^\pi B_0\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Finite-batch measurements remain diagnostics; they certify neither the uniform saturation and Lipschitz assumptions nor the required sup-norm residual.

B.1 OPTIONAL FIXED-POINT CPI SUBSTITUTION

Corollary B.4 (One-proposal parameter dependence). *Under Corollaries 5.2 and 5.3 and Theorem 7.3,*

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{1-\gamma} \left(\frac{\varepsilon_{\text{res}}^*}{1-\gamma^K} + L_V \frac{L_z^n}{1-L_z} C_z \right) - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (47)$$

This is the optional fixed-point specialization of the overlap-aware finite-reference penalty $2\alpha\gamma\tau_\infty B_{n,m}/(1-\gamma)$. For $n \geq 1$, its recurrent term vanishes as $L_z \rightarrow 0$; together with $\varepsilon_{\text{res}}^* \rightarrow 0$, this recovers the exact-mixture CPI expression. At $n = 0$, the recurrent term instead tends to $L_V C_z$.

A projection-aware version substitutes $\varepsilon_0 + (1+\gamma^K)L_V(R)R$ for $\varepsilon_{\text{res}}^*$ and $\bar{L}_z^{\text{sat}}(R)$ for L_z ; the full expression is Eq. (148) in Appendix K.5. Local finite differences certify neither uniform factor.

For $\pi_{\text{new}} = (1-\alpha)\pi + \alpha\pi_{\text{cand}}$:

$$\begin{aligned} \eta(\pi_{\text{new}}) &\geq L_\pi(\pi_{\text{new}}) - \frac{\gamma\alpha^2\Delta_g}{(1-\gamma)(1-\gamma+\gamma\alpha)} \\ &\geq L_\pi(\pi_{\text{new}}) - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}, \end{aligned} \quad (48)$$

where g , Δ_g , and ε_{CPI} use the $\mathcal{C}_{\text{mix}}$ -restricted definitions in Theorem 7.3.

$$\pi_{\text{new}} = (1-\alpha)\pi + \alpha\pi_{\text{cand}}, \quad \alpha \in [0, 1]. \quad (49)$$

C PERSISTENT- z ON THE AUGMENTED MDP

C.1 DIRECT FINITE-DEPTH RESIDUAL CERTIFICATE

The formal statement below applies Theorem 5.1 on a measurable augmented domain. In particular, it permits a continuous carried latent $z \in \mathbb{R}^d$ without reducing it to a countable reachable set.

Fix n , (θ, ψ) , the policy parameters, and all environment kernels. Let $h \in \{1, 2, \dots, H\}$ be the number of edits remaining before the current action. Define

$$F_n(x, y, z) := T_{x,y}^{\circ n}(z) = \tilde{f}_\theta^{\circ n}(z, y, x), \quad \bar{s} = (x, y, z, h), \quad (50)$$

For a nonabsorbing state, first compute F_n and sample an action. If the environment transition terminates for any reason (including a solution or terminal STOP), or if $h = 1$, the successor is the shared atom s_{abs} ; budget exhaustion is terminal. Otherwise the clock-complete successor is

$$\bar{s}' = (x, \text{edit}(y, a; x), F_n(x, y, z), h-1).$$

Thus the state stores the pre-unroll carried latent z , the policy uses the post-unroll latent $F_n(x, y, z)$, and a nonterminal successor carries that post-unroll latent. The current and candidate action kernels share this same frozen F_n and hence the same augmented MDP:

$$\bar{\pi}(a \mid \bar{s}) = \pi_\phi(a \mid y, F_n(x, y, z), x), \quad \bar{\pi}_{\text{cand}}(a \mid \bar{s}) = \pi_{\text{cand}}(a \mid y, F_n(x, y, z), x).$$

Changing the recurrent map between these two policies would change the transition kernel and is outside ordinary CPI.

Remark C.1 (Why the recurrent map is shared). A small Euclidean discrepancy between deterministic recurrent maps does not imply a small policy or augmented-kernel discrepancy. For example, take $F_\varepsilon(z) = -\varepsilon/2$ and $G_\varepsilon(z) = \varepsilon/2$ and a fixed threshold policy head that chooses disjoint actions below and above zero. Then $\sup_z \|F_\varepsilon(z) - G_\varepsilon(z)\| = \varepsilon$, but the action laws are disjoint and a fixed MDP can have an order- $1/(1-\gamma)$ return gap. Moreover, $\text{TV}(\delta_{F(z)}, \delta_{G(z)}) = 1$ whenever $F(z) \neq G(z)$. A different-map theorem therefore needs explicit reward and transition-kernel discrepancies and sufficient regularity to propagate them. Map distance alone is insufficient.

The initial law is

$$\bar{\rho}(dx, dy, dz, dh) = \rho(dx, dy, dh) \delta_{z_{\text{init}}(x,y)}(dz), \quad (51)$$

or, for stochastic initialization, $\bar{\rho}(dx, dy, dz, dh) = \rho(dx, dy, dh)\nu_0(dz | x, y, h)$ with any recorded measurable initialization kernel. Ignoring h is an architectural specialization, not a Bellman requirement.

Assume the initialization, edit/reward kernels, F_n , V_ψ , and both policy kernels are measurable, and the K -step expected reward and evaluator are bounded. Let $\bar{\mathcal{C}}_\pi$ be a measurable $\bar{\pi}$ -invariant domain with $\bar{\rho}(\bar{\mathcal{C}}_\pi) = 1$. Let $\bar{\mathcal{C}}$ be a measurable advantage-evaluation domain containing $\bar{\mathcal{C}}_\pi$ and the shared absorber, containing the one-step successors from $\bar{\mathcal{C}}_\pi$ for $\bar{\pi}$ - and $\bar{\pi}_{\text{cand}}$ -almost every action, and invariant under every subsequent $\bar{\pi}$ step. In particular, $\bar{\rho}(\bar{\mathcal{C}}) = 1$. Write $\bar{\mathcal{C}}^\circ := \bar{\mathcal{C}} \setminus \{s_{\text{abs}}\}$ for its nonabsorbing part. Keep this augmented MDP, including its deployed depth n , fixed. For any finite comparison depth $q \geq 0$, define the endpoint function

$$\tilde{U}_q(\bar{s}) := V_\psi(\tilde{f}_\theta^{\circ q}(z, y, x), x, y), \quad \tilde{U}_q(s_{\text{abs}}) := -C_{\text{max}}. \quad (52)$$

Changing q here changes only a comparison function on the already-fixed state space. It does not change F_n , either policy kernel, or the augmented transition kernel. Without this restriction, two endpoint depths would belong to different MDPs and no finite-reference comparison would follow. At $q = n$, Eq. (52) is the deployed evaluator. Let $\tilde{\mathcal{T}}_K^\pi$ denote the K -step Bellman operator of this frozen augmented MDP. Its direct residual is

$$\bar{\varepsilon}_{\text{res},n} := \|\tilde{U}_n - \tilde{\mathcal{T}}_K^\pi \tilde{U}_n\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}. \quad (53)$$

Here $\eta(\bar{\pi}')$ denotes the discounted return of any augmented policy $\bar{\pi}'$ from $\bar{\rho}$, and $d_{\bar{\pi}}$ is the discounted occupancy of the current policy. For an augmented advantage estimator $\hat{A}$, define

$$\hat{L}_{\bar{\pi}}(\bar{\pi}') := \eta(\bar{\pi}) + \frac{1}{1-\gamma} \mathbb{E}_{\bar{s} \sim d_{\bar{\pi}}, a \sim \bar{\pi}'(\cdot | \bar{s})} [\hat{A}(\bar{s}, a)]. \quad (54)$$

Corollary C.2 (Direct persistent- z certificate on the augmented MDP). *Let $\bar{\mathcal{C}}$ be the preceding measurable augmented domain and equip $B_b(\bar{\mathcal{C}})$ with its ordinary pointwise sup norm. Under the preceding frozen-MDP, measurability, bounded- K -step-reward, common-boundary, and invariance assumptions, take $\tilde{U}_n \in B_b(\bar{\mathcal{C}})$. Theorem 5.1 gives a unique $V^\pi \in B_b(\bar{\mathcal{C}})$ and*

$$\|\tilde{U}_n - V^\pi\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \frac{\bar{\varepsilon}_{\text{res},n}}{1-\gamma^K}. \quad (55)$$

If $\hat{A}_{\tilde{U}_n}$ uses the exact transition expectation and exact baseline and is therefore centered pointwise under $\bar{\pi}$ on the full augmented state, then Assumption 7.1 holds for every $\bar{s} \in \bar{\mathcal{C}}_\pi$ and for $\bar{\pi}$ - and $\bar{\pi}_{\text{cand}}$ -almost every action, with the valid choice

$$\bar{\varepsilon}_A := \frac{2\gamma \bar{\varepsilon}_{\text{res},n}}{1-\gamma^K}. \quad (56)$$

If, additionally, the analyzed policy is the pointwise exact mixture $\bar{\pi}_{\text{new}} = (1-\alpha)\bar{\pi} + \alpha\bar{\pi}_{\text{cand}}$ with no parameter interpolation or distillation, let $\bar{\mathcal{C}}_{\text{mix}}$ be a measurable augmented domain carrying $\bar{\rho}$, containing $\bar{\mathcal{C}}_\pi$, and invariant under both $\bar{\pi}$ and $\bar{\pi}_{\text{new}}$. Define

$$\bar{\varepsilon}_{\text{CPI}} := \sup_{\bar{s} \in \bar{\mathcal{C}}_{\text{mix}}} |\mathbb{E}_{a \sim \bar{\pi}_{\text{cand}}(\cdot | \bar{s})} A^\pi(\bar{s}, a)|, \quad (57)$$

$$\bar{\tau}_\infty := \sup_{\bar{s} \in \bar{\mathcal{C}}_\pi} \text{TV}(\bar{\pi}_{\text{cand}}(\cdot | \bar{s}), \bar{\pi}(\cdot | \bar{s})), \quad (58)$$

where $\bar{\mathcal{C}}_\pi \subseteq \bar{\mathcal{C}} \cap \bar{\mathcal{C}}_{\text{mix}}$ is the common full-measure current-policy domain. Then Theorem 7.3 gives

$$\eta(\bar{\pi}_{\text{new}}) \geq \hat{L}_{\bar{\pi}}(\bar{\pi}_{\text{new}}) - \frac{2\alpha\gamma\bar{\tau}_\infty}{1-\gamma} \frac{\bar{\varepsilon}_{\text{res},n}}{1-\gamma^K} - \frac{2\bar{\varepsilon}_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (59)$$

For the target-network analogue, let $\bar{r}_K^\pi$ be the K -step expected reward of this same fixed augmented MDP. Suppose the conditional target trajectory follows $\bar{\pi}$, and take $\tilde{U}_q, \bar{V} \in B_b(\bar{\mathcal{C}})$ with the common scalar absorbing boundary. Define

$$\bar{\varepsilon}_{\text{tar},q} := \left\| \tilde{U}_q - (\bar{r}_K^\pi + \gamma^K P_\pi^K \bar{V}) \right\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}, \quad (60)$$

$$\bar{R}_{\text{tar},q}^P := \bar{\varepsilon}_{\text{tar},q} + \gamma^K \|P_\pi^K(\bar{V} - \tilde{U}_q)\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}, \quad (61)$$

$$\bar{R}_{\text{tar},q}^{\text{sup}} := \bar{\varepsilon}_{\text{tar},q} + \gamma^K \|\bar{V} - \tilde{U}_q\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}. \quad (62)$$

Then

$$\|\tilde{U}_q - \tilde{T}_K^{\bar{\pi}} \tilde{U}_q\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \bar{R}_{\text{targ}, q}^P \leq \bar{R}_{\text{targ}, q}^{\text{sup}}, \quad (63)$$

$$\|\tilde{U}_q - V^{\bar{\pi}}\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \frac{\bar{R}_{\text{targ}, q}^P}{1 - \gamma^K}. \quad (64)$$

For fixed-MDP endpoint depths $0 \leq n < m$,

$$\|\tilde{U}_n - V^{\bar{\pi}}\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \|\tilde{U}_n - \tilde{U}_m\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} + \frac{\bar{R}_{\text{targ}, m}^P}{1 - \gamma^K}. \quad (65)$$

Let $\hat{A}_{\tilde{U}_q}$ be the one-step estimator of Eq. (145) on the augmented MDP, with its transition expectation and current-policy baseline evaluated exactly conditional on the full augmented state. Define

$$\bar{\varepsilon}_{A, \text{cand}} := \sup_{\bar{s} \in \bar{\mathcal{C}}_{\pi}} \left| \mathbb{E}_{a \sim \bar{\pi}_{\text{cand}}(\cdot | \bar{s})} \left[\hat{A}_{\tilde{U}_q}(\bar{s}, a) - A^{\bar{\pi}}(\bar{s}, a) \right] \right|. \quad (66)$$

Exact baseline evaluation centers this estimator conditional on the full augmented state, and the candidate bias satisfies

$$\bar{\varepsilon}_{A, \text{cand}} \leq 2\gamma\bar{\tau}_{\infty} \frac{\bar{R}_{\text{targ}, q}^P}{1 - \gamma^K}. \quad (67)$$

The target-network CPI and finite-reference CPI forms are therefore

$$\eta(\bar{\pi}_{\text{new}}) \geq \hat{L}_{\bar{\pi}}(\bar{\pi}_{\text{new}}) - \frac{2\alpha\gamma\bar{\tau}_{\infty}}{(1 - \gamma)(1 - \gamma^K)} \bar{R}_{\text{targ}, q}^P - \frac{2\bar{\varepsilon}_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}, \quad (68)$$

$$\eta(\bar{\pi}_{\text{new}}) \geq \hat{L}_{\bar{\pi}}(\bar{\pi}_{\text{new}}) - \frac{2\alpha\gamma\bar{\tau}_{\infty}}{1 - \gamma} \left(\|\tilde{U}_n - \tilde{U}_m\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} + \frac{\bar{R}_{\text{targ}, m}^P}{1 - \gamma^K} \right) - \frac{2\bar{\varepsilon}_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (69)$$

The last display uses the same exact value-derived construction at $\tilde{U}_n$. Every target formula keeps the propagated lag before the ordinary sup-lag fallback.

This direct result is Bellman contraction on the actual finite-depth augmented evaluator. It requires no recurrent latent fixed point, recurrent contraction, Lipschitz head, projection, or slow-drift assumption; it still uses the value-function fixed point $V^{\bar{\pi}}$. The augmented residual differs from the episodic residual on (x, y, h) because both the policy and transition can depend on the carried latent. A learned instantiation would additionally need to establish the uniform augmented residual, exact augmented-state centering, and pointwise exact-mixture deployment. Empirical regression loss, a finite-batch maximum, and population L^2 error establish neither the target population sup norm nor either required target-lag sup norm.

Remark C.3 (Scope of the direct augmented-state certificate). Corollary C.2 is a result on $\bar{s} = (x, y, z, h)$, not on the plan marginal (x, y) . Its exact-centering condition averages actions conditional on the full $\bar{s}$, and its mixture identity must hold pointwise on the augmented mixture-occupancy domain $\bar{\mathcal{C}}_{\text{mix}}$. Finite-batch maxima are diagnostics, not uniform certificates.

Lemma C.4 (Finite-path carried-latent comparison). Let $z_{t+1} = F_t(z_t)$ for $t = 0, \dots, T-1$, and let $\xi_0, \dots, \xi_T$ be an arbitrary reference path. Suppose along these paths that, for every $t = 0, \dots, T-1$,

$$\|F_t(z_t) - F_t(\xi_t)\| \leq \kappa_t \|z_t - \xi_t\|, \quad d_t := \|F_t(\xi_t) - \xi_{t+1}\|.$$

Then, with empty products equal to one,

$$\|z_t - \xi_t\| \leq \|z_0 - \xi_0\| \prod_{i=0}^{t-1} \kappa_i + \sum_{j=0}^{t-1} d_j \prod_{i=j+1}^{t-1} \kappa_i. \quad (70)$$

If a head evaluated directly at z_t is $L_{V,t}$ -Lipschitz, its error is at most $L_{V,t}$ times the right-hand side. In the architecture used here, the state stores the pre-unroll z_t and the head reads $F_t(z_t)$; relative to $F_t(\xi_t)$ its error is therefore at most $L_{V,t}\kappa_t\|z_t - \xi_t\|$.

Proof. Writing $e_t = \|z_t - \xi_t\|$, the triangle inequality gives $e_{t+1} \leq \kappa_t e_t + d_t$. Induction gives Eq. (70); head Lipschitzness gives the final claims. This is a pathwise statement. It becomes a uniform sup-norm certificate only when the initial error, κ_t , d_t , and $L_{V,t}$ are bounded uniformly over every declared nonabsorbing start and trajectory. $\square$

C.2 OPTIONAL SLOW-DRIFT COMPARISON TO A FIXED-POINT REFERENCE

Slow drift is needed only for the stronger architectural comparison between the actual transient evaluator and the memoryless reference $\tilde{U}_*(x, y, z, h) := V_\psi(z^*(x, y), x, y)$, with $\tilde{U}_*(s_{\text{abs}}) = -C_{\text{max}}$. For this subsection, invoke Assumption B.1 with the declared domain $\mathcal{D}^\circ = \bar{\mathcal{C}}^\circ$. Thus $z^*(x, y)$ denotes the unique fixed point only for input pairs represented by this augmented closure; no claim is made about other pairs. Assume $(\bar{s}, u) \mapsto \tilde{f}_\theta(u, y(\bar{s}), x(\bar{s}))$ is jointly measurable on $\bar{\mathcal{C}}^\circ \times \mathcal{Z}_{\text{inv}}$ and impose a measurable iteration initialization $z^{(0)} : \bar{\mathcal{C}}^\circ \rightarrow \mathcal{Z}_{\text{inv}}$. Proposition B.2 then makes the fixed-point map measurable. Invoke Assumption B.3 with $\mathcal{D}^\circ = \bar{\mathcal{C}}^\circ$ and $\mathcal{Z}_{\text{bd}} = \mathcal{Z}_{\text{inv}}$; with the separate finite absorbing boundary, $\tilde{U}_*$ is bounded and measurable on the augmented closure. Assume the tracking conditions below hold uniformly over nonabsorbing trajectory segments formed by one action drawn under either policy followed by $\bar{\pi}$ dynamics, and assume every carried latent on those segments lies in $\mathcal{Z}_{\text{inv}}$. These segments include $\bar{\rho}$ -almost every initial augmented state and every starting state used to generate a one-deviation trajectory in the augmented closure. In the stochastic-initialization case, require $\nu_0(\mathcal{Z}_{\text{inv}} \mid x, y, h) = 1$ for ρ -almost every (x, y, h) . Let $e : \mathcal{Y} \rightarrow \mathcal{E}_y$ be a fixed plan embedding and write $\|y - y'\|_\mathcal{E} := \|e(y) - e(y')\|_{\mathcal{E}_y}$.

Assumption C.5 (Slow drift of fixed points). Uniformly over the nonabsorbing augmented trajectory segments declared above, there exist $L_{z^*} \geq 0$, $\Delta y_{\text{max}} \geq 0$, and $E_0 < \infty$ such that

1. for every consecutive source/successor pair with common instance x and represented plans y_t, y_{t+1} , $\|z^*(x, y_t) - z^*(x, y_{t+1})\| \leq L_{z^*} \|y_t - y_{t+1}\|_\mathcal{E}$;
2. every admissible edit satisfies $\|y_{t+1} - y_t\|_\mathcal{E} \leq \Delta y_{\text{max}}$; and
3. for $\bar{\rho}$ -almost every (x, y_0, z_0, h_0) , and uniformly for every augmented state treated as time zero when constructing a one-deviation trajectory in the policy-pair closure, the carried latent satisfies $\|z_0 - z^*(x, y_0)\| \leq E_0$.

Remark C.6 (A conditional sufficient bound for L_{z^*}). If y has a specified differentiable continuous representation, f_θ is C^1 near an interior fixed point, and $\|\partial_z f_\theta\|_{\text{op}} \leq L_z < 1$, the implicit-function calculation gives the local derivative bound

$$\|D_y z^*(x, y)\|_{\text{op}} \leq \frac{\|D_y f_\theta(z^*(x, y), y, x)\|_{\text{op}}}{1 - L_z}. \quad (71)$$

This pointwise statement does not by itself supply the single uniform L_{z^*} required by Assumption C.5. For every represented consecutive plan pair, that conclusion additionally needs a C^1 fixed-point branch along a connecting path and one derivative bound uniform over all such paths. For discrete Sudoku plans, the manuscript provides and verifies no such extension.

Remark C.7 (Discrete-plan scope). For a discrete domain, Assumption C.5 is simply an additional regularity assumption in the stated embedding. Successful training does not by itself establish it, and finite-batch diagnostics do not establish the uniform assumption.

Remark C.8 (Scope of the slow-drift assumption). Only the augmented closure for the frozen policy pair is relevant; no global smoothness claim outside that closure is needed.

For this optional subsection fix $n \geq 1$. At nonabsorbing times define $e_t = \|z_t - z^*(x, y_t)\|$ and $\kappa_n = L_z^n < 1$.

Lemma C.9 (Drift bound for persistent- z). *Under the $\mathcal{D}^\circ = \bar{\mathcal{C}}^\circ$ invocation of Assumption B.1 and Assumption C.5, every transition for which both $\bar{s}_t$ and $\bar{s}_{t+1}$ are nonabsorbing satisfies*

$$e_{t+1} \leq \kappa_n e_t + L_{z^*} \Delta y_{\text{max}}, \quad (72)$$

and

$$\sup_{t: \bar{s}_t \neq s_{\text{abs}}} \|z_t^{(n)} - z^*(x, y_t)\| \leq C_{\text{drift}}(n) := \kappa_n \max\left(E_0, \frac{L_{z^*} \Delta y_{\text{max}}}{1 - \kappa_n}\right) \leq \kappa_n E_0 + \frac{\kappa_n L_{z^*} \Delta y_{\text{max}}}{1 - \kappa_n}. \quad (73)$$

Lemma C.9 is the uniform specialization of Lemma C.4 with $F_t = T_{x, y_t}^{\circ n}$, $\xi_t = z^*(x, y_t)$, $\kappa_t = L_z^n$, and $d_t \leq L_{z^*} \Delta y_{\text{max}}$. The additional factor κ_n in $C_{\text{drift}}(n)$ appears because the value head reads the post-unroll latent.

Table 2: Dependency map for the main result families.

Result	Required beyond the fixed snapshot	Not required
Finite-reference value, Theorem 5.1	Bellman contraction on an invariant domain and bounded finite endpoints; head Lipschitzness only for the latent-path upper bound	Recurrent contraction, projection, slow drift, policy mixing
Weighted fixed-policy restatement, Proposition E.6	Domination of the one-policy K -skeleton occupancy by the chosen reference measure	Policy-sequence concentrability, recurrent contraction, finite-sample inference
Target-network bridge, Proposition 6.1	Population target residual and propagated target lag on the same Bellman domain	Recurrent convergence, finite-sample inference
Signed CPI, Theorem 7.3	Exact probability mixture, mixture invariance, advantage span, and signed defect control	Recurrent contraction or projection
Persistent augmented state, Corollary C.2	Clock-complete Markovization and one shared frozen F_n	Recurrent contraction, projection, slow drift
Optional fixed-point and drift results	Recurrent contraction and an invariant latent set; projection only when used to establish that set; slow drift only for the persistent memoryless comparison	None of these assumptions enter the primary finite-reference result
Deployment, Theorem H.5	Recursive deployment-domain invariance and uniform TV, or either-direction KL via Pinsker	Exact equality of the realized policy and mixture

Remark C.10 (When the drift bound is informative). The bound is useful only when its assumed constants are finite and calibrated on the policy-pair closure. Otherwise the direct residual certificate remains mathematically valid but uncalibrated, while the fixed-point comparison is vacuous.

Define

$$\bar{\varepsilon}_{\text{res},*} := \|\tilde{U}_* - \bar{T}_K^{\bar{\pi}} \tilde{U}_*\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}. \quad (74)$$

Then Lipschitzness of the head and Lemma C.9 give

$$\|\tilde{U}_n - \tilde{U}_*\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq L_V C_{\text{drift}}(n), \quad (75)$$

$$\|\tilde{U}_n - V^{\bar{\pi}}\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq L_V C_{\text{drift}}(n) + \frac{\bar{\varepsilon}_{\text{res},*}}{1 - \gamma^K}, \quad (76)$$

$$|\bar{\varepsilon}_{\text{res},n} - \bar{\varepsilon}_{\text{res},*}| \leq (1 + \gamma^K) L_V C_{\text{drift}}(n). \quad (77)$$

These are optional comparisons to a memoryless fixed-point reference; none is needed for Corollary C.2.

D NOTATION SUMMARY

This appendix collects the notation and result dependencies used in the main text. The dependency table separates the finite-endpoint result from its optional contraction and finite-horizon refinements.

We summarize notation here for convenience. Table 3 lists additional symbols used in proofs.

E CONSTRUCTIVE DEFINITION OF THE ADVANTAGE-EVALUATION CLOSURE

Different parts of the argument follow different paths. Value-to-advantage control needs one candidate action followed by the base policy; CPI must also account for repeated mixture decisions. The table summarizes these requirements before the formal definitions.

Definition E.1 (π -invariant Bellman domain). A measurable set $\mathcal{C}_\pi$ is a Bellman/value domain for π if $\rho(\mathcal{C}_\pi) = 1$ whenever the initial law is used and

$$P(\mathcal{C}_\pi \mid s, a) = 1 \quad \text{for } \pi(da \mid s)\text{-almost every } a \text{ and every } s \in \mathcal{C}_\pi.$$

Equivalently, the policy-induced kernel leaves $\mathcal{C}_\pi$ invariant. Theorem 5.1 needs only such a domain and does not need a candidate policy.

Definition E.2 (Advantage-evaluation domain). Fix current and candidate stochastic kernels π and π_{cand} . A measurable set $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ contains a π -invariant domain $\mathcal{C}_\pi$ and the common absorber and satisfies

1. for every $s \in \mathcal{C}_\pi$, $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} \mid s, a) = 1$ for $\pi(da \mid s)$ - and $\pi_{\text{cand}}(da \mid s)$ -almost every action; and

Table 3: Additional symbols and notations

Symbol	Meaning
$h, \bar{s} = (x, y, z, h)$	Remaining edit budget / clock-complete persistent state
$\mathcal{D}^\circ$	Declared nonabsorbing domain for a contraction invocation
$\mathcal{C}, \mathcal{C}^\circ$	Value domain / nonabsorbing latent domain
P_{rec}	Declared recurrent projection operator: Π_R for $R > 0$, or $\text{Id}_{\mathcal{Z}}$
$\mathcal{Z}_{\text{bd}}, V_{\text{max}}$	Declared bounded-head latent set / its uniform finite bound
$z^{(j)}(s), s \in \mathcal{C}^\circ$	Latent after finite depth j
U_j, U_*	Finite-depth value / optional fixed-point value, with scalar boundary
$B_{n,m}$	Finite path/discrepancy plus depth- m residual certificate
$\tilde{U}_n(\bar{s}), \tilde{U}_*(\bar{s})$	Persistent finite-depth value / memoryless reference
$\varepsilon_{\text{res},n}$	Episodic finite-depth Bellman residual
$\varepsilon_{\text{targ},q}$	Target-network conditional population residual at endpoint depth q
$R_{\text{targ},q}^P, R_{\text{targ},q}^{\text{sup}}$	Propagated-lag target bound / ordinary sup-lag fallback
$d_{\rho}^{\pi,K}$	Discounted occupancy of the fixed K -step current-policy skeleton
$C_{\rho,\mu}^{\pi,K}, C_{\rho,\mu}^{(0)}$	Fixed-kernel occupancy / initial-law density-ratio bounds for the weighted- L_p restatement
$\bar{\varepsilon}_{\text{res},n}$	Persistent augmented-state finite-depth residual
$\bar{\varepsilon}_{\text{targ},q}$	Persistent target-network population residual at endpoint depth q
$\bar{R}_{\text{targ},q}^P, \bar{R}_{\text{targ},q}^{\text{sup}}$	Persistent propagated-lag target bound / sup-lag fallback
$\bar{\varepsilon}_{\text{res},*}$	Augmented residual of the fixed-point reference
$L_{z^*}, \Delta y_{\text{max}}, E_0$	Slow-drift parameters
$\varepsilon_A, \bar{\varepsilon}_A$	Episodic / augmented advantage error bounds
$\varepsilon_{A,\text{cand}}, \bar{\varepsilon}_{A,\text{cand}}$	Episodic / augmented candidate-defect suprema
$\varepsilon_{\text{res}}^*$	Architectural residual at U_*
g, Δ_g	Candidate expected advantage / span on the declared mixture or pair domain
$\hat{g}, \beta_d, M_d$	Estimated candidate slope / one-sided signed-defect bound / certified slope
$\mathcal{C}_{\text{pair}}$	Common π/π_{cand} -invariant domain for step intervals
$c, b(s), \Xi_\alpha$	Signed centering and candidate estimator defects
$\tau(s), \delta_{\text{dep}}$	Policy overlap / deployment TV radius
ε_{CPI}	Uniform scalar bound on $ g $
$\kappa_n = L_z^n$	Contraction after n steps
$B_V(s)$	Baseline: $\mathbb{E}_{b \sim \pi}[\hat{Q}_V(s, b)]$

Table 4: Domain and theorem map. Every domain carries the stated initial law. The superscript $^\circ$ removes the absorber only for latent-path, fixed-point, and head-Lipschitz assumptions; value norms include the absorber.

Domain	Required invariance	Paths included	$\alpha?$	Used by
$\mathcal{C}_\pi$	P_π	Repeated current-policy decisions	No	Bellman values, d_π , centering
$\mathcal{R}_{\pi,\pi_{\text{cand}}}^{\text{adv}}$	P_π after the first action	One current or candidate deviation, then π	No	Finite-reference and value-to-advantage bounds
$\mathcal{C}_{\text{mix}}$	P_π and P_{π_α}	Repeated exact-mixture decisions	May	Theorem 7.3
$\mathcal{C}_{\text{pair}}$	P_π and $P_{\pi_{\text{cand}}}$	Repeated decisions for every exact mixture weight	No	Step-interval corollary
$\mathcal{D}_{\text{dep}}$	P_{π_α} and $P_{\bar{\pi}}$	Repeated reference-mixture and deployed-policy decisions	Through the selected π_α	Theorem H.5
Barred counterparts	The corresponding augmented kernels with one shared F_n	The same closures on (x, y, z, h)	As above	Persistent-state results

2. for every $s \in \mathcal{R}_{\pi,\pi_{\text{cand}}}^{\text{adv}}$, $P(\mathcal{R}_{\pi,\pi_{\text{cand}}}^{\text{adv}} \mid s, a) = 1$ for $\pi(da \mid s)$ -almost every action.

Thus the domain contains current-policy evaluation states, one current or candidate action, and every subsequent current-policy continuation. We write $\|f\|_{\infty,\pi,\pi_{\text{cand}}}^{\text{adv}}$ for the ordinary supremum on this set, including the scalar absorbing boundary.

Definition E.3 (Mixture-occupancy and deployment domains). For a fixed α , a measurable $\mathcal{C}_{\text{mix}}$ is a mixture-occupancy domain if $\rho(\mathcal{C}_{\text{mix}}) = 1$ and it is invariant under both π and $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$. It therefore supports the state laws at every time under the current policy and under repeated mixture decisions, not merely a one-deviation path. A deployment domain $\mathcal{D}_{\text{dep}}$ satisfies $\rho(\mathcal{D}_{\text{dep}}) = 1$ and is recursively invariant under both π_α and $\tilde{\pi}$. The CPI span theorem uses $\mathcal{C}_{\text{mix}}$; the deployment theorem uses $\mathcal{D}_{\text{dep}}$.

For a certificate simultaneous over more than one mixture weight, a measurable common policy-pair domain $\mathcal{C}_{\text{pair}}$ satisfies

$$\rho(\mathcal{C}_{\text{pair}}) = 1, \quad P_\pi(\mathcal{C}_{\text{pair}} \mid s) = P_{\pi_{\text{cand}}}(\mathcal{C}_{\text{pair}} \mid s) = 1 \quad \text{for every } s \in \mathcal{C}_{\text{pair}}. \quad (78)$$

By linearity of policy-induced kernels, every exact mixture π_α also leaves $\mathcal{C}_{\text{pair}}$ invariant. Thus $\mathcal{C}_{\text{pair}}$ is a valid single choice of $\mathcal{C}_{\text{mix}}$ for every $\alpha \in [0, 1]$.

Remark E.4 (Recovering candidate invariance from a positive mixture). Let $\alpha_0 > 0$. If a measurable set $\mathcal{E}$ is invariant under π and π_{α_0} , then for every $s \in \mathcal{E}$,

$$1 = P_{\pi_{\alpha_0}}(\mathcal{E} \mid s) = (1 - \alpha_0)P_\pi(\mathcal{E} \mid s) + \alpha_0 P_{\pi_{\text{cand}}}(\mathcal{E} \mid s),$$

and $P_\pi(\mathcal{E} \mid s) = 1$ forces $P_{\pi_{\text{cand}}}(\mathcal{E} \mid s) = 1$. This implication does not follow at $\alpha_0 = 0$ without separately assuming candidate invariance.

In finite or countable spaces these domains can be constructed with the usual positive-probability reachable sets and action supports. On arbitrary measurable action spaces we use the policy-almost-every-action clauses above; no topological action support is presumed.

Why boundedness matters. On the countable zero-reward chain $s_i \mapsto s_{i+1}$ with one action and $\gamma \in (0, 1)$, add an unreachable zero-reward absorber with $U(s_{\text{abs}}) = V^\pi(s_{\text{abs}}) = 0$. The unbounded function $U(s_i) = \gamma^{-i}$ has zero K -step residual for every $K \geq 1$, while its sup-norm error from $V^\pi = 0$ is infinite. This is why the residual theorem works on bounded functions, not arbitrary pointwise solutions.

Remark E.5 (Why the primary extension uses B_b). Theorem 5.1 works directly on $B_b(\mathcal{C})$ and needs no dominating measure. An alternative formulation on equivalence classes in a common $L^\infty(\nu)$ would additionally need initial and occupancy domination by ν , ν -nonsingularity of the relevant kernels, representative-independent Bellman integration, and the corresponding joint measurability conditions. Occupancy comparisons for general measures use total variation; vector ℓ_1 notation is reserved for finite or countable spaces.

E.1 A WEIGHTED- L_p FIXED-POLICY RESTATEMENT

A weighted norm can replace the population supremum when a reference measure covers the fixed policy's discounted occupancy. The following result states the corresponding change-of-measure cost. For a probability measure ν and $1 \leq p < \infty$, write $\|f\|_{p,\nu} := (\int_{\mathcal{C}} |f|^p d\nu)^{1/p}$.

Proposition E.6 (Weighted- L_p finite-reference bound). *Under the measurable fixed-policy hypotheses of Theorem 5.1, let ρ and μ be probability measures on $\mathcal{C}$, let $1 \leq p < \infty$, and take bounded measurable endpoints U_n, U_m . Set*

$$\beta := \gamma^K, \quad P := P_\pi^K, \quad r_m := U_m - \mathcal{T}_K^\pi U_m, \quad (79)$$

and define the discounted occupancy of the K -step skeleton by

$$d_\rho^{\pi,K} := (1 - \beta) \sum_{t=0}^{\infty} \beta^t \rho P^t. \quad (80)$$

This is a probability measure on $\mathcal{C}$. Suppose $d_\rho^{\pi,K} \ll \mu$ and

$$C_{\rho,\mu}^{\pi,K} := \left\| \frac{dd_\rho^{\pi,K}}{d\mu} \right\|_{\infty,\mu} < \infty, \quad (81)$$

where the norm is the μ -essential supremum. Then

$$\|U_n - V^\pi\|_{p,\rho} \leq \|U_n - U_m\|_{p,\rho} + \frac{(C_{\rho,\mu}^{\pi,K})^{1/p}}{1 - \gamma^K} \|U_m - \mathcal{T}_K^\pi U_m\|_{p,\mu}. \quad (82)$$

If also $\rho \ll \mu$ and

$$C_{\rho,\mu}^{(0)} := \left\| \frac{d\rho}{d\mu} \right\|_{\infty,\mu} < \infty, \quad (83)$$

then the endpoint discrepancy may instead be measured under μ :

$$\|U_n - V^\pi\|_{p,\rho} \leq (C_{\rho,\mu}^{(0)})^{1/p} \|U_n - U_m\|_{p,\mu} + \frac{(C_{\rho,\mu}^{\pi,K})^{1/p}}{1 - \gamma^K} \|r_m\|_{p,\mu}. \quad (84)$$

Proof. Let $e_m := U_m - V^\pi$. Subtracting the fixed-point identity $V^\pi = \mathcal{T}_K^\pi V^\pi$ from the definition of r_m gives

$$e_m = r_m + \beta P e_m. \quad (85)$$

Because e_m and r_m are bounded and $\beta < 1$, iteration of this identity and uniform decay of the remainder yield

$$e_m = \sum_{t=0}^{\infty} \beta^t P^t r_m, \quad |e_m| \leq \sum_{t=0}^{\infty} \beta^t P^t |r_m|. \quad (86)$$

Define the Markov kernel $Q := (1 - \beta) \sum_{t \geq 0} \beta^t P^t$. Then $\rho Q = d_\rho^{\pi,K}$ and $|e_m| \leq Q|r_m|/(1 - \beta)$. Jensen's inequality for the Markov kernel Q gives

$$\begin{aligned} \|e_m\|_{p,\rho}^p &\leq \frac{1}{(1 - \beta)^p} \int_{\mathcal{C}} Q|r_m|^p d\rho \\ &= \frac{1}{(1 - \beta)^p} \int_{\mathcal{C}} |r_m|^p dd_\rho^{\pi,K} \\ &\leq \frac{C_{\rho,\mu}^{\pi,K}}{(1 - \beta)^p} \|r_m\|_{p,\mu}^p. \end{aligned} \quad (87)$$

Taking p th roots and applying the triangle inequality proves Eq. (82). Under Eq. (83), the same change-of-measure calculation gives $\|U_n - U_m\|_{p,\rho} \leq (C_{\rho,\mu}^{(0)})^{1/p} \|U_n - U_m\|_{p,\mu}$, which proves Eq. (84). $\square$

The contraction and resolvent in Proposition E.6 are generated by the Bellman operator $\mathcal{T}_K^\pi$ and the fixed K -step kernel P_π^K . The learned recurrence is not a replacement Bellman operator. It determines the endpoint functions and, in persistent mode, is part of the frozen augmented transition kernel. The coefficient in Eq. (81) concerns this one fixed kernel; it does not include a supremum over changing policy sequences, a function-class inherent Bellman error, or a sampling term. Thus the proposition is an a posteriori fixed-snapshot analogue, not the weighted-norm API/FVI results of Munos (Munos, 2003) or Munos and Szepesvári (Munos & Szepesvári, 2008).

Table 5: Positioning against primary approximate-DP and CPI sources. Entries describe theorem objects and assumptions, not empirical rankings.

Work	Main object and sequence	Error, coverage, and statistical terms	Scope relative to this paper
Munos (2003) API	Recursive approximate evaluation and policy improvement	Weighted quadratic evaluation errors under source-specific state weighting	Policy-loss error propagation; no finite recurrent endpoint or persistent latent
Munos–Szepesvári (2008) FVI	Sequence of sampling-based fitted value updates and a resulting policy	Weighted L_p bounds, discounted-average concentrability, inherent Bellman residual, and finite-sample terms	No finite recurrent endpoint or persistent latent
Kakade–Langford (2002) CPI and Kakade (2003)	Recursively promoted conservative policy mixtures	Restart-distribution and approximate-greedy-oracle conditions	Improvement, termination, and approximate optimality under source assumptions; no recurrent endpoint
Recurrent reasoning models	Finite latent computation or equilibrium representation	Predictor- or fixed-point-specific objectives	No plan-MDP persistent-state or exact-mixture CPI guarantee
This work	One fixed π , candidate, snapshot, and endpoint pair	Population sup norm; no statistical bridge or mismatch coefficient in the primary theorem	Finite n, m and persistent augmented state; no policy-sequence, finite-sample, or near-optimality claim

Domination can be especially restrictive for persistent deterministic latents. Augmented occupancies may lie on lower-dimensional graphs, so a generic product reference measure with Lebesgue measure on the latent can assign those graphs measure zero and fail to dominate the occupancy. Choosing μ to dominate the actual occupancy can make the coefficient finite, and $\mu = d_p^{\pi, K}$ makes it one, but the latter is an analytical identity, not a practical regression-distribution guarantee. An exogenous training distribution may have poor or singular coverage. Finite reachable-set or finite-population diagnostics do not establish the continuous-state population coefficient.

F EXACT FINITE-MDP DERIVATION

This section derives Table 1. It uses exact rational arithmetic and no data, learned parameter, or random sample. The state and action definitions are

State	Action	(r, S')
s_0	a	$(0, s_0)$
s_0	b	$(1, s_1)$
s_1	a	$(0, s_0)$
s_1	b	$(0, s_1)$
s_{abs}	a or b	$(0, s_{\text{abs}})$

The current policy is $\pi(a | s) = 1$, the candidate is $\pi_{\text{cand}}(b | s) = 1$, and $\rho = \delta_{s_0}$. These definitions also apply at the absorber, where both actions induce the same zero-reward self-loop. Thus

$$\mathcal{C}_\pi = \{s_0, s_{\text{abs}}\}, \quad \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} = \mathcal{C}_{\text{mix}} = \mathcal{C}_{\text{pair}} = \mathcal{D}_{\text{dep}} = \mathcal{S}.$$

The first set is π -invariant and carries ρ . One candidate deviation from s_0 reaches s_1 , after which π returns to s_0 . Repeated mixture decisions remain in $\mathcal{S}$, so this is the complete relevant closure.

Finite-reference value calculation. Set $\gamma = 1/2$, $K = 1$, $n = 1$, and $m = 2$. On the scalar latent space, use the common initialization $z^{(0)}(s) = 0$, head $V_\psi(z, s) = z$, and

$$T_s(z) = u_s - \frac{1}{2}z, \quad (u_{s_0}, u_{s_1}, u_{s_{\text{abs}}}) = \left(\frac{1}{4}, -\frac{1}{4}, 0\right).$$

The absorber therefore has the common depth-independent boundary zero. The finite endpoints are

$$U_1 = \left(\frac{1}{4}, -\frac{1}{4}, 0\right), \quad U_2 = \left(\frac{1}{8}, -\frac{1}{8}, 0\right).$$

Under π , all nonabsorbing rewards are zero and both nonabsorbing states move to s_0 , so $V^\pi = (0, 0, 0)$. Moreover,

$$\mathcal{T}_1^\pi U_2 = \left(\frac{1}{16}, \frac{1}{16}, 0\right), \quad U_2 - \mathcal{T}_1^\pi U_2 = \left(\frac{1}{16}, -\frac{3}{16}, 0\right).$$

Consequently,

$$\begin{aligned} \|U_1 - U_2\|_\infty &= \frac{1}{8}, & \|U_2 - \mathcal{T}_1^\pi U_2\|_\infty &= \frac{3}{16}, \\ \|U_1 - V^\pi\|_\infty &\leq \frac{1}{8} + \frac{3/16}{1 - 1/2} = \frac{1}{2}, & \|U_1 - V^\pi\|_\infty &= \frac{1}{4}. \end{aligned}$$

The certificate gives the finite error radius $1/2$ without a recurrent fixed-point argument. It is not tight: the actual error is smaller by a factor of two.

Exact mixture and occupancies. Take $\alpha = 1/2$. At every state, $\pi_{1/2}(a \mid s) = \pi_{1/2}(b \mid s) = 1/2$. On the two nonabsorbing states,

$$P_{\pi_{1/2}} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}, \quad r_{\pi_{1/2}} = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}.$$

The absorber is a separate zero-reward self-loop. Solving $d = (1 - \gamma)\rho + \gamma dP$ gives

$$d_\pi = (1, 0, 0), \quad d_{\pi_{1/2}} = \left(\frac{3}{4}, \frac{1}{4}, 0\right).$$

Solving the mixture Bellman equations gives $V^{\pi_{1/2}} = (3/4, 1/4, 0)$ and hence $\eta(\pi_{1/2}) = 3/4$.

Advantage, centering, and CPI lower bound. Because $V^\pi = 0$, the candidate expected advantage is

$$g = (1, 0, 0), \quad \Delta_g = 1.$$

For the value-derived one-step estimator based on U_1 , the exact action values on (s_0, s_1) are

$$\hat{Q}_{U_1} = \begin{pmatrix} 1/8 & 7/8 \\ 1/8 & -1/8 \end{pmatrix},$$

with columns (a, b) . Exact centering under the deterministic current policy subtracts the first column. Including the absorber gives

$$\hat{A}(a) = (0, 0, 0), \quad \hat{A}(b) = \left(\frac{3}{4}, -\frac{1}{4}, 0\right).$$

Thus the centering defect and candidate defect from Eq. (24) are

$$c = (0, 0, 0), \quad b = \left(-\frac{1}{4}, -\frac{1}{4}, 0\right), \quad \Xi_{1/2} = -\frac{1}{8}.$$

The estimated surrogate is

$$\hat{L}_\pi(\pi_{1/2}) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi_{1/2}} \hat{A}(s, a) = 2 \left(\frac{1}{2} \cdot \frac{3}{4}\right) = \frac{3}{4}.$$

The exact span penalty is

$$\frac{\gamma \alpha^2 \Delta_g}{(1 - \gamma)(1 - \gamma + \gamma \alpha)} = \frac{1}{3}.$$

Equation (27) therefore yields

$$\eta(\pi_{1/2}) \geq \frac{3}{4} - \frac{-1/8}{1/2} - \frac{1}{3} = \frac{2}{3},$$

while the exact return is $3/4$. In contrast, the absolute uniform specialization uses $\varepsilon_{A,\text{cand}} = 1/4$ and $\varepsilon_{\text{CPI}} = 1$, producing

$$\frac{3}{4} - \frac{1}{4} - \frac{2}{3} = -\frac{1}{6}.$$

This valid but vacuous bound shows why the signed defect and the advantage span matter. Run `python3 UPI_TRM_ICLR/verify_finite_mdp.py` from the repository root to reproduce the calculation.

G SHAPING CONVENTION AND ABSORBING-STATE NORMALIZATION

We use potential-based shaping with potential $\Phi(s) = c(x, y)$ for episodic $s = (x, y, h)$ and persistent $\bar{s} = (x, y, z, h)$ (the potential ignores the clock and latent), and $\Phi(s_{\text{abs}}) = C_{\text{max}}$ on their shared absorbing atom. We define

$$r(s, a, s') = r_0(s, a, s') + \gamma \Phi(s') - \Phi(s), \quad (88)$$

applying shaping on *all* transitions, including the absorbing self-loop.

For the absorbing self-loop, taking $r_0(s_{\text{abs}}, a, s_{\text{abs}}) = 0$ gives

$$r(s_{\text{abs}}, a, s_{\text{abs}}) = \gamma \Phi(s_{\text{abs}}) - \Phi(s_{\text{abs}}) = (\gamma - 1)C_{\text{max}}.$$

Thus every policy satisfies the fixed shaped boundary value

$$V^\pi(s_{\text{abs}}) = \sum_{t=0}^{\infty} \gamma^t (\gamma - 1)C_{\text{max}} = -C_{\text{max}}.$$

If one prefers the standard normalization $V(s_{\text{abs}}) = 0$, use the shifted potential $\tilde{\Phi}(s) := \Phi(s) - C_{\text{max}}$. This adds the constant reward $(1 - \gamma)C_{\text{max}}$ to every transition and shifts all returns by $+C_{\text{max}}$ independently of the policy. Accordingly, performance differences and optimal policies are unchanged.

H FINITE-HORIZON AND DEPLOYMENT-AWARE BOUNDS

The remaining edit budget permits a stagewise analysis. Residual errors are propagated only until the common terminal boundary, replacing infinite geometric amplification by a finite sum. The CPI and deployment results below likewise count only decisions within that horizon.

This section uses the finite edit clock directly. Fix $0 \leq \gamma < 1$, an integer horizon $H \geq 0$, and one shared absorbing atom with boundary value

$$b := V_h^\pi(s_{\text{abs}}) = -C_{\text{max}} \quad \text{for every } h, \quad r_{\text{abs}} = (1 - \gamma)b.$$

Thus $b = r_{\text{abs}} + \gamma b$. Every reward interval below includes r_{abs} .

For each $h \in \{0, \dots, H\}$, let $\mathcal{C}_h$ be a measurable stage domain containing s_{abs} , with

$$\mathcal{C}_h^\circ := \mathcal{C}_h \setminus \{s_{\text{abs}}\}, \quad \mathcal{C}_0 = \{s_{\text{abs}}\}.$$

Value norms use $\|f\|_{\infty, \mathcal{C}_h} := \sup_{s \in \mathcal{C}_h} |f(s)|$; latent paths are defined only on $\mathcal{C}_h^\circ$.

Let $Q_h(dr, ds' | s, a)$ be the measurable controlled reward-transition kernel from stage h to stage $h - 1$, let P_h be its state marginal, and let π_h be the current-policy stochastic kernel. Assume its induced kernel satisfies $P_h^\pi(\mathcal{C}_{h-1} | s) = 1$ for every $s \in \mathcal{C}_h$. If an initial law is used, assume $\rho(\mathcal{C}_H) = 1$. Assume the rewards and conditional expected rewards below are bounded and measurable. For $1 \leq \ell \leq h$ and $W \in B_b(\mathcal{C}_{h-\ell})$ with $W(s_{\text{abs}}) = b$, define

$$(\mathcal{T}_{h:\ell}^\pi W)(s) := \mathbb{E} \left[\sum_{i=0}^{\ell-1} \gamma^i r_i + \gamma^\ell W(S_\ell) \mid S_0 = s, A_i \sim \pi_{h-i}(\cdot | S_i) \right], \quad (89)$$

where an absorbed trajectory remains at the shared atom. Padding after early termination with absorbing self-loops and applying b at the end of the block account for the absorbing tail exactly once. These are current-policy Bellman/value domains; the repeated-mixture domains used for CPI are defined separately below.

Theorem H.1 (Measurable finite-horizon residual and finite-reference bound). Fix $K \geq 1$. Let $U_{h,q} \in B_b(C_h)$ for every $h \in \{0, \dots, H\}$ and every finite depth $q \geq 0$, and impose

$$U_{h,q}(s_{\text{abs}}) = b, \quad U_{0,q} \equiv b \quad \text{for every finite } q \geq 0. \quad (90)$$

The true stage value satisfies $V_0^\pi \equiv b$ and $V_h^\pi(s_{\text{abs}}) = b$. For $h \geq 1$, set

$$\begin{aligned} \ell_h &:= \min\{K, h\}, & J_h &:= \left\lceil \frac{h}{K} \right\rceil, \\ \epsilon_{h,q} &:= \|U_{h,q} - \mathcal{T}_{h:\ell_h}^\pi U_{h-\ell_h,q}\|_{\infty, C_h}, & D_{h;n,m} &:= \|U_{h,n} - U_{h,m}\|_{\infty, C_h}, \end{aligned} \quad (91)$$

Then, for finite depths $0 \leq n < m$,

$$\|U_{h,n} - V_h^\pi\|_{\infty, C_h} \leq D_{h;n,m} + \sum_{j=0}^{J_h-1} \gamma^{jK} \epsilon_{h-jK,m}. \quad (92)$$

Consequently, with $\epsilon_m^{(h)} := \max_{1 \leq r \leq h} \epsilon_{r,m}$,

$$\|U_{h,n} - V_h^\pi\|_{\infty, C_h} \leq D_{h;n,m} + \frac{1 - \gamma^{KJ_h}}{1 - \gamma^K} \epsilon_m^{(h)}. \quad (93)$$

Taking the same finite depth as the reference gives

$$\|U_{h,q} - V_h^\pi\|_{\infty, C_h} \leq \sum_{j=0}^{J_h-1} \gamma^{jK} \epsilon_{h-jK,q}. \quad (94)$$

For the recurrent specialization, fix one measurable frozen recurrent map F , a measurable initialization on each C_h° , a latent norm, and a bounded measurable scalar head V_ψ . Define only on C_h°

$$z_h^{(q+1)}(s) = F(z_h^{(q)}(s), s), \quad U_{h,q}(s) = V_\psi(z_h^{(q)}(s), x(s), y(s)).$$

For persistent state, $z_h^{(0)}(x, y, z, h) = z$ is the carried pre-unroll latent. An initialization that ignores h is an architectural specialization, not a premise of the residual argument. At the absorber use Eq. (90); no absorbing latent is defined. If the head is uniformly L_V -Lipschitz in the latent argument, then

$$D_{h;n,m} \leq L_V \sup_{s \in C_h^\circ} \|z_h^{(n)}(s) - z_h^{(m)}(s)\| \quad (95)$$

$$\leq L_V \sup_{s \in C_h^\circ} \sum_{q=n}^{m-1} \|z_h^{(q+1)}(s) - z_h^{(q)}(s)\| \quad (96)$$

$$\leq L_V \sum_{q=n}^{m-1} \sup_{s \in C_h^\circ} \|z_h^{(q+1)}(s) - z_h^{(q)}(s)\|. \quad (97)$$

For $h = 0$, set $J_0 = 0$, $D_{0;n,m} = 0$, and interpret every residual and path sum as empty. If C_h° is empty, the nonnegative latent suprema in the recurrent clause are defined to be zero.

Proof. For a fixed depth q , write $E_{h,q} := \|U_{h,q} - V_h^\pi\|_{\infty, C_h}$ and $E_{0,q} = 0$. The stage Bellman recursion and the Markov-kernel expectation inequality give

$$\begin{aligned} E_{h,q} &\leq \epsilon_{h,q} + \|\mathcal{T}_{h:\ell_h}^\pi U_{h-\ell_h,q} - \mathcal{T}_{h:\ell_h}^\pi V_{h-\ell_h}^\pi\|_{\infty, C_h} \\ &\leq \epsilon_{h,q} + \gamma^{\ell_h} E_{h-\ell_h,q}. \end{aligned}$$

On trajectories absorbed inside the block, the terminal-function difference is zero because both functions equal b . Unrolling visits stages $h, h-K, \dots, h-(J_h-1)K$; the last partial block ends at $E_{0,q} = 0$. This proves Eq. (94). Adding and subtracting $U_{h,m}$ proves Eq. (92); the uniform residual bound gives Eq. (93). Finally, head Lipschitzness gives Eq. (95); the triangle inequality for the pointwise telescoping path gives Eq. (96), and sup-sum interchange gives Eq. (97). $\square$

In a persistent-state application, the state is (x, y, z, h) . Whenever the current and candidate policies are analyzed in one augmented MDP, both use the same frozen recurrent map for action selection and for the carried post-unroll latent on each nonterminal transition.

For finite-horizon policy improvement, define

$$\eta_H(\pi) := \mathbb{E} \left[\sum_{t=0}^{H-1} \gamma^t r_t + \gamma^H b \right],$$

where early terminal trajectories are padded by the absorbing self-loop. Let d_t^π be the state law before decision t , including absorbed mass. For each h , let $\mathcal{M}_h$ be a measurable stagewise mixture-occupancy domain containing s_{abs} , with $\mathcal{M}_0 = \{s_{\text{abs}}\}$ and $\rho(\mathcal{M}_H) = 1$. Require the controlled successor kernel to satisfy

$$P_h(\mathcal{M}_{h-1} \mid s, a) = 1$$

for every $s \in \mathcal{M}_h$ and for $\pi_h(da \mid s)$ - and $\pi_{\text{cand},h}(da \mid s)$ -almost every action. This recursive invariance implies $d_{H-h}^\pi(\mathcal{M}_h) = d_{H-h}^{\pi_\alpha}(\mathcal{M}_h) = 1$ and covers a candidate first step followed by every current-policy continuation. No topological action support is presumed. Define every stage advantage to be zero at the absorbing atom.

Theorem H.2 (Finite-horizon CPI with stagewise spans). *Fix bounded measurable stage rewards and the common terminal boundary b . For every $h \in \{1, \dots, H\}$, fix a current-policy kernel π_h , a candidate kernel $\pi_{\text{cand},h}$, and the exact pointwise probability-space mixture*

$$\pi_{\alpha,h}(\cdot \mid s) = (1 - \alpha)\pi_h(\cdot \mid s) + \alpha\pi_{\text{cand},h}(\cdot \mid s), \quad \alpha \in [0, 1]. \quad (98)$$

All policies act in the same fixed time-indexed MDP. Assume the relevant values, advantages, and estimator functions are bounded and measurable on $\mathcal{M}_h$. Define

$$g_h(s) := \mathbb{E}_{a \sim \pi_{\text{cand},h}(\cdot \mid s)}[A_h^\pi(s, a)], \quad \Delta_h := \sup_{s \in \mathcal{M}_h} g_h(s) - \inf_{s \in \mathcal{M}_h} g_h(s), \quad (99)$$

with $g_h(s_{\text{abs}}) = 0$. Let $\hat{A}_h$ be exactly centered:

$$\mathbb{E}_{a \sim \pi_h(\cdot \mid s)}[\hat{A}_h(s, a)] = 0 \quad \text{for every } s \in \mathcal{M}_h, \quad (100)$$

and define the signed candidate defect

$$\beta_h(s) := \mathbb{E}_{a \sim \pi_{\text{cand},h}(\cdot \mid s)}[\hat{A}_h(s, a) - A_h^\pi(s, a)]. \quad (101)$$

Set

$$\Xi_{H,\alpha} := \alpha \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{s \sim d_t^\pi}[\beta_{H-t}(s)]. \quad (102)$$

Define

$$\hat{L}_{H,\pi}(\pi_\alpha) := \eta_H(\pi) + \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{\substack{s \sim d_t^\pi, \\ a \sim \pi_{\alpha,H-t}(\cdot \mid s)}}[\hat{A}_{H-t}(s, a)]. \quad (103)$$

Then

$$\begin{aligned} \eta_H(\pi_\alpha) &\geq \hat{L}_{H,\pi}(\pi_\alpha) - \Xi_{H,\alpha} \\ &\quad - \alpha \sum_{t=0}^{H-1} \gamma^t \Delta_{H-t} [1 - (1 - \alpha)^t]. \end{aligned} \quad (104)$$

Because $\Xi_{H,\alpha}$ is signed, any established upper bound $B_{H,\alpha} \geq \Xi_{H,\alpha}$ may replace it. One valid choice is

$$B_{H,\alpha} := \alpha \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{s \sim d_t^\pi} |\beta_{H-t}(s)|. \quad (105)$$

Proof. The time-indexed performance-difference identity is

$$\eta_H(\pi_\alpha) - \eta_H(\pi) = \alpha \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{s \sim d_t^{\pi_\alpha}} [g_{H-t}(s)]. \quad (106)$$

The common terminal term cancels. Represent each mixture decision by an independent Bernoulli selector. Before decision t , the first t selectors have all chosen the current-policy component with probability $(1 - \alpha)^t$. Conditional on that event the state law is d_t^π , so for some probability law ν_t ,

$$d_t^{\pi_\alpha} = (1 - \alpha)^t d_t^\pi + [1 - (1 - \alpha)^t] \nu_t.$$

Therefore

$$\text{TV}(d_t^{\pi_\alpha}, d_t^\pi) \leq 1 - (1 - \alpha)^t. \quad (107)$$

The exponent is t , not $t + 1$, because d_t is measured before decision t . This selector proof needs no maximal-coupling kernel. Applying $|\mathbb{E}_\mu f - \mathbb{E}_\nu f| \leq \text{TV}(\mu, \nu) \text{span}(f)$ to g_{H-t} bounds the occupancy term. Exact centering gives

$$\widehat{L}_{H,\pi}(\pi_\alpha) - \eta_H(\pi) = \alpha \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{s \sim d_t^\pi} [g_{H-t}(s) + \beta_{H-t}(s)].$$

Subtracting proves Eq. (104); moving absolute values inside the current-occupancy expectation proves Eq. (105). $\square$

Corollary H.3 (Uniform finite-horizon CPI specialization). *Under Theorem H.2, define*

$$\varepsilon_{\text{CPI},h} := \sup_{s \in \mathcal{M}_h} |g_h(s)|, \quad \varepsilon_{A,\text{cand},h} := \sup_{s \in \mathcal{M}_h} |\beta_h(s)|. \quad (108)$$

Then $\Delta_h \leq 2\varepsilon_{\text{CPI},h}$ and

$$\begin{aligned} \eta_H(\pi_\alpha) &\geq \widehat{L}_{H,\pi}(\pi_\alpha) - \alpha \sum_{t=0}^{H-1} \gamma^t \varepsilon_{A,\text{cand},H-t} \\ &\quad - 2\alpha \sum_{t=0}^{H-1} \gamma^t \varepsilon_{\text{CPI},H-t} [1 - (1 - \alpha)^t]. \end{aligned} \quad (109)$$

For uniform bounds $\varepsilon_{A,\text{cand},h} \leq \varepsilon_{A,\text{cand}}$ and $\varepsilon_{\text{CPI},h} \leq \varepsilon_{\text{CPI}}$, define

$$G_H(q) := \sum_{t=0}^{H-1} q^t = \frac{1 - q^H}{1 - q}, \quad 0 \leq q < 1. \quad (110)$$

The total penalty in Eq. (109) is at most

$$\alpha \varepsilon_{A,\text{cand}} G_H(\gamma) + 2\alpha \varepsilon_{\text{CPI}} [G_H(\gamma) - G_H(\gamma(1 - \alpha))]. \quad (111)$$

Corollary H.4 (Stagewise policy-overlap refinement). *Fix a finite evaluator depth q and bounded measurable approximations $U_{h,q}$ on the mixture domains. Write*

$$E_h := \|U_{h,q} - V_h^\pi\|_{\infty, \mathcal{M}_h}, \quad E_0 = 0,$$

where $U_{0,q} = V_0^\pi = b$ on $\mathcal{M}_0 = \{s_{\text{abs}}\}$. Define

$$\tau_h(s) := \text{TV}(\pi_{\text{cand},h}(\cdot | s), \pi_h(\cdot | s)), \quad \tau_{h,\infty} := \sup_{s \in \mathcal{M}_h} \tau_h(s).$$

Suppose $\widehat{A}_h$ is the exactly centered one-step estimator derived from $U_{h-1,q}$, with reward and transition expectations evaluated exactly. Then

$$|\beta_h(s)| \leq 2\gamma E_{h-1} \tau_h(s), \quad (112)$$

$$\varepsilon_{A,\text{cand},h} \leq 2\gamma E_{h-1} \tau_{h,\infty}, \quad (113)$$

$$\mathbb{E}_{s \sim d_{H-h}^\pi} |\beta_h(s)| \leq 2\gamma E_{h-1} \mathbb{E}_{s \sim d_{H-h}^\pi} \tau_h(s). \quad (114)$$

In particular, the common terminal boundary gives $E_0 = 0$, so the stage-1 advantage-estimation defect is zero.

More generally, let $\ell_h = \min\{K, h\}$ and define the partial-block estimator

$$\hat{Q}_{h,q}^{(\ell_h)}(s, a) := \mathbb{E} \left[\sum_{i=0}^{\ell_h-1} \gamma^i r_i + \gamma^{\ell_h} U_{h-\ell_h,q}(S_{\ell_h}) \mid \begin{array}{l} S_0=s, A_0=a; \\ A_i \sim \pi_{h-i}(\cdot | S_i), 1 \leq i < \ell_h \end{array} \right], \quad (115)$$

and center it exactly under π_h . Its candidate defect satisfies

$$|\beta_h^{(\ell_h)}(s)| \leq 2\gamma^{\ell_h} E_{h-\ell_h} \tau_h(s), \quad (116)$$

$$\varepsilon_{A,\text{cand},h}^{(\ell_h)} \leq 2\gamma^{\ell_h} E_{h-\ell_h} \tau_{h,\infty}. \quad (117)$$

Thus a final partial block with $K > h$ has $\ell_h = h$ and zero bootstrap error because $E_0 = 0$.

If the residual theorem is used to certify E_h , its Bellman domains must contain the mixture domains $\mathcal{M}_h$ and satisfy the corresponding invariance assumptions. A one-deviation closure alone is insufficient. Under those enlarged-domain premises, take the estimator depth $q = n < m$. Equation (92) gives

$$\varepsilon_{A,\text{cand},h} \leq 2\gamma \tau_{h,\infty} \left[D_{h-1;n,m} + \sum_{j=0}^{J_{h-1}-1} \gamma^{jK} \epsilon_{h-1-jK,m} \right], \quad (118)$$

and Eq. (94) gives

$$\varepsilon_{A,\text{cand},h} \leq 2\gamma \tau_{h,\infty} \sum_{j=0}^{J_{h-1}-1} \gamma^{jK} \epsilon_{h-1-jK,n}. \quad (119)$$

For an ℓ_h -step estimator, the corresponding explicit bounds are

$$\varepsilon_{A,\text{cand},h}^{(\ell_h)} \leq 2\gamma^{\ell_h} \tau_{h,\infty} \left[D_{h-\ell_h;n,m} + \sum_{j=0}^{J_{h-\ell_h}-1} \gamma^{jK} \epsilon_{h-\ell_h-jK,m} \right], \quad (120)$$

$$\varepsilon_{A,\text{cand},h}^{(\ell_h)} \leq 2\gamma^{\ell_h} \tau_{h,\infty} \sum_{j=0}^{J_{h-\ell_h}-1} \gamma^{jK} \epsilon_{h-\ell_h-jK,n}. \quad (121)$$

The fallback follows from $\tau_h \leq 1$.

Proof. Let $e_{h-1} = U_{h-1,q} - V_{h-1}^\pi$ and $q_e(s, a) = \gamma \mathbb{E}[e_{h-1}(S_1) \mid s, a]$. Exact centering gives $\beta_h = \mathbb{E}_{\pi_{\text{cand},h}} q_e - \mathbb{E}_{\pi_h} q_e$. Recursive invariance gives a common action set of full current- and candidate-policy measure on which $q_e(s, a) \in [-\gamma E_{h-1}, \gamma E_{h-1}]$. The TV expectation inequality on this essential range proves Eqs. (112)–(114). For the partial block, the reward terms cancel and leave $q_e^{(\ell_h)}(s, a) = \gamma^{\ell_h} \mathbb{E}[e_{h-\ell_h}(S_{\ell_h}) \mid s, a]$, which lies in $[-\gamma^{\ell_h} E_{h-\ell_h}, \gamma^{\ell_h} E_{h-\ell_h}]$ on the analogous common essential action range. Substituting the two finite-horizon residual bounds proves the final displays. $\square$

For $H = 0$, $\eta_0(\pi) = b$, $G_0 = 0$, and every sum and penalty is empty. At $h = 0$, values equal the common boundary and no advantage is needed; set $g_0 = \Delta_0 = 0$ if convenient. When $K > h$, the last backup is the partial block $\ell_h = h$ and its terminal error vanishes. When $\gamma = 0$, only the $t = 0$ reward remains and the occupancy-shift penalty is zero. At $\alpha = 0$ every policy-shift term vanishes; at $\alpha = 1$ the selector bound is zero at $t = 0$ and one for $t \geq 1$. A nonzero common boundary cancels exactly in every policy comparison. In a folded zero-bootstrap representation, the terminal tail is already included in the folded reward and must not be added again.

Theorem H.5 (First-mismatch deployment perturbation). *Fix $0 \leq \gamma < 1$ and $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$ for $\alpha \in [0, 1]$. Let $\tilde{\pi}$ be a deployed stochastic kernel in the same fixed MDP as π_α , with the same initial law ρ . Assume the state and action spaces are standard Borel, and let $\mathsf{K}(dr, ds' \mid s, a)$ be the measurable joint reward-transition kernel. The policy kernels then admit a measurable statewise maximal-coupling kernel.*

Let $\mathcal{D}_{\text{dep}}$ be measurable with $\rho(\mathcal{D}_{\text{dep}}) = 1$, and assume the policy-induced transition kernels of both π_α and $\tilde{\pi}$ leave $\mathcal{D}_{\text{dep}}$ invariant. For every $s \in \mathcal{D}_{\text{dep}}$, assume K assigns rewards in $[r_{\min}, r_{\max}]$ for $\pi_\alpha(da | s)$ - and $\tilde{\pi}(da | s)$ -almost every action. Write $\Delta_r := r_{\max} - r_{\min}$ and suppose

$$\sup_{s \in \mathcal{D}_{\text{dep}}} \text{TV}(\tilde{\pi}(\cdot | s), \pi_\alpha(\cdot | s)) \leq \delta_{\text{dep}}, \quad \delta_{\text{dep}} \in [0, 1]. \quad (122)$$

Then

$$|\eta(\tilde{\pi}) - \eta(\pi_\alpha)| \leq \frac{\Delta_r \delta_{\text{dep}}}{(1 - \gamma)(1 - \gamma + \gamma \delta_{\text{dep}})}. \quad (123)$$

The extra denominator factor satisfies $1 - \gamma + \gamma \delta_{\text{dep}} \geq 1 - \gamma$, so the first-mismatch constant improves the generic perturbation constant.

For an integer horizon $H \geq 0$, impose the same assumptions on the clock-complete deployment domain at every remaining horizon. If both returns use the same scalar terminal boundary b , then

$$\begin{aligned} |\eta_H(\tilde{\pi}) - \eta_H(\pi_\alpha)| &\leq \Delta_r \sum_{t=0}^{H-1} \gamma^t [1 - (1 - \delta_{\text{dep}})^{t+1}] \\ &= \Delta_r [G_H(\gamma) - (1 - \delta_{\text{dep}})G_H(\gamma(1 - \delta_{\text{dep}}))]. \end{aligned} \quad (124)$$

The common term $\gamma^H b$ cancels exactly, including when $b \neq 0$.

Proof. Couple the two processes from the same initial state. Whenever their states agree at decision t , use a measurable maximal coupling of $\pi_\alpha(\cdot | s_t)$ and $\tilde{\pi}(\cdot | s_t)$. Conditional on a matched history, their actions agree with probability at least $1 - \delta_{\text{dep}}$. When the actions agree, draw one shared reward and successor state from $K(\cdot | s_t, a_t)$. After the first action mismatch, use any measurable coupling.

Let E_t be the event that the actions agree at every decision $0, \dots, t$. Iterated conditioning gives

$$\Pr(E_t) \geq (1 - \delta_{\text{dep}})^{t+1}.$$

The exponent is $t + 1$ because the reward at time t follows decision t . On E_t , the two rewards at time t are identical. Since both rewards lie in an interval of width Δ_r ,

$$|\mathbb{E}[\tilde{r}_t] - \mathbb{E}[r_t^\alpha]| \leq \Delta_r \Pr(E_t^c) \leq \Delta_r [1 - (1 - \delta_{\text{dep}})^{t+1}].$$

Summing proves the finite-horizon bound; the common terminal scalar cancels. Letting $H \rightarrow \infty$ is valid because rewards are bounded, and

$$\begin{aligned} \sum_{t=0}^{\infty} \gamma^t [1 - (1 - \delta_{\text{dep}})^{t+1}] &= \frac{1}{1 - \gamma} - \frac{1 - \delta_{\text{dep}}}{1 - \gamma(1 - \delta_{\text{dep}})} \\ &= \frac{\delta_{\text{dep}}}{(1 - \gamma)(1 - \gamma + \gamma \delta_{\text{dep}})}. \end{aligned} \quad \square$$

Remark H.6 (Attainment by a first-mismatch process). The constants in Theorem H.5 are attained by a two-state MDP. Let ℓ be initial and u absorbing. At ℓ , action a_0 gives reward 0 and returns to ℓ , while a_1 gives reward Δ_r and enters u ; every action at u gives reward Δ_r and remains there. Let π_α choose a_0 at ℓ , and let $\tilde{\pi}$ choose a_1 with probability δ_{dep} and a_0 otherwise. Let the policies agree at u . The reward difference at time t is exactly $\Delta_r [1 - (1 - \delta_{\text{dep}})^{t+1}]$, so equality holds in both bounds, with terminal boundary $b = 0$ in the finite-horizon construction.

Corollary H.7 (Either-direction uniform-KL deployment bound). *Under Theorem H.5, suppose either*

$$\begin{aligned} \sup_{s \in \mathcal{D}_{\text{dep}}} D_{\text{KL}}(\tilde{\pi}(\cdot | s) \| \pi_\alpha(\cdot | s)) &\leq \kappa, \quad \text{or} \\ \sup_{s \in \mathcal{D}_{\text{dep}}} D_{\text{KL}}(\pi_\alpha(\cdot | s) \| \tilde{\pi}(\cdot | s)) &\leq \kappa \end{aligned}$$

for a finite κ . Define

$$\bar{\delta}_\kappa := \min \left\{ 1, \sqrt{\frac{\kappa}{2}} \right\}. \quad (125)$$

Then

$$|\eta(\tilde{\pi}) - \eta(\pi_\alpha)| \leq \frac{\Delta_r \bar{\delta}_\kappa}{(1 - \gamma)(1 - \gamma + \gamma \bar{\delta}_\kappa)}, \quad (126)$$

and

$$|\eta_H(\tilde{\pi}) - \eta_H(\pi_\alpha)| \leq \Delta_r [G_H(\gamma) - (1 - \bar{\delta}_\kappa)G_H(\gamma(1 - \bar{\delta}_\kappa))]. \quad (127)$$

Proof. Pinsker's inequality in either direction and $\text{TV} \leq 1$ show that the actual uniform TV discrepancy is at most $\bar{\delta}_\kappa$. Apply Theorem H.5 with $\delta_{\text{dep}} = \bar{\delta}_\kappa$; both right-hand sides are increasing in this radius. If the selected KL divergence is infinite, this corollary does not apply; the TV theorem remains available when Eq. (122) is established directly. $\square$

Corollary H.8 (Occupancy-aware CPI with deployment discrepancy). *Assume Theorems 7.3 and H.5 hold in one fixed MDP with one initial law. Let $\mathcal{C}_{\text{mix}}$ satisfy Definition E.3, and let $\mathcal{D}_{\text{dep}}$ be measurable, carry the same initial law, and be recursively invariant under both π_α and $\tilde{\pi}$. Assume the standard-Borel, bounded-reward, and uniform-TV conditions of Theorem H.5. For any established one-sided bound $B_\alpha \geq \Xi_\alpha$,*

$$\begin{aligned} \eta(\tilde{\pi}) \geq & \hat{L}_\pi(\pi_\alpha) - \frac{B_\alpha}{1 - \gamma} - \frac{\gamma \alpha^2 \Delta_g}{(1 - \gamma)(1 - \gamma + \gamma \alpha)} \\ & - \frac{\Delta_r \delta_{\text{dep}}}{(1 - \gamma)(1 - \gamma + \gamma \delta_{\text{dep}})}. \end{aligned} \quad (128)$$

A distilled, logit-interpolated, hidden-state-interpolated, parameter-interpolated, or otherwise realized policy is covered only when it acts in this same MDP and its TV or KL discrepancy from the exact probability-space mixture is established uniformly on $\mathcal{D}_{\text{dep}}$. For persistent state (x, y, z, h) , all policies must use the same frozen recurrent map that determines the carried-latent transition. Changing that map changes the MDP.

Proof. Lower-bound $\eta(\tilde{\pi})$ by $\eta(\pi_\alpha)$ minus Eq. (123), then apply Eq. (27) with $\Xi_\alpha \leq B_\alpha$. $\square$

Remark H.9 (Boundary cases and uniform scope). At $\delta_{\text{dep}} = 0$, both deployment penalties are zero. At $\delta_{\text{dep}} = 1$, they are $\Delta_r/(1 - \gamma)$ and $\Delta_r G_H(\gamma)$. At $\gamma = 0$ and $H \geq 1$, both reduce to $\Delta_r \delta_{\text{dep}}$; at $H = 0$, the finite-horizon penalty is zero.

The residual, centering, TV, and KL premises are uniform on their declared domains. Replay-batch means, batch maxima, held-out finite maxima, and local finite-difference proxies are diagnostics. They do not establish the uniform premises.

I PROOF OF THE OCCUPANCY-AWARE CPI RESULT

The proof first corrects the estimated surrogate by its signed error, then bounds the change from base to mixture occupancy. It uses the performance-difference identity (Kakade & Langford, 2002) and the Bernoulli selector calculation in Lemma 7.2.

Proof of Theorem 7.3. For the true surrogate

$$L_\pi(\pi') := \eta(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi'(\cdot|s)} [A^\pi(s, a)],$$

linearity of the exact mixture and $\mathbb{E}_{a \sim \pi(\cdot|s)} A^\pi(s, a) = 0$ give

$$\begin{aligned} \hat{L}_\pi(\pi_\alpha) - L_\pi(\pi_\alpha) &= \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi} [(1 - \alpha)c(s) + \alpha b(s)] \\ &= \frac{\Xi_\alpha}{1 - \gamma}, \end{aligned}$$

which proves Eq. (26).

The performance-difference identity also gives

$$\eta(\pi_\alpha) - L_\pi(\pi_\alpha) = \frac{\alpha}{1 - \gamma} (\mathbb{E}_{d_{\pi_\alpha}} g - \mathbb{E}_{d_\pi} g).$$

For any bounded measurable g and probability measures p, q , $|\mathbb{E}_p g - \mathbb{E}_q g| \leq \text{span}(g) \text{TV}(p, q)$. Applying this inequality on $\mathcal{C}_{\text{mix}}$ and then Lemma 7.2 yields

$$\eta(\pi_\alpha) - L_\pi(\pi_\alpha) \geq -\frac{\gamma\alpha^2\Delta_g}{(1-\gamma)(1-\gamma+\gamma\alpha)}.$$

Combining the last display with the exact surrogate identity proves Eq. (27). Any $B_\alpha \geq \Xi_\alpha$ may replace the signed term. Under exact centering, $c = 0$ and $\Xi_\alpha = \alpha\mathbb{E}_{d_\pi} b \leq \alpha\varepsilon_{A,\text{cand}}$. Finally, $\Delta_g \leq 2\sup_s |g(s)| = 2\varepsilon_{\text{CPI}}$, proving Eq. (31) and Eq. (48). $\square$

Corollary I.1 (Occupancy-averaged centering and candidate defects). *Under Theorem 7.3, let*

$$\varepsilon_{\text{cent},d} := \mathbb{E}_{s \sim d_\pi} |c(s)|, \quad \varepsilon_{A,\text{cand},d} := \mathbb{E}_{s \sim d_\pi} |b(s)|.$$

Then

$$\eta(\pi_\alpha) \geq \widehat{L}_\pi(\pi_\alpha) - \frac{(1-\alpha)\varepsilon_{\text{cent},d} + \alpha\varepsilon_{A,\text{cand},d}}{1-\gamma} - \frac{\gamma\alpha^2\Delta_g}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (129)$$

Pointwise uniform bounds on $|c|$ and $|b|$ imply the corresponding looser version immediately. For a value-derived estimator, Eq. (14) supplies $\varepsilon_{A,\text{cand},d} \leq 2\gamma\delta_V \mathbb{E}_{d_\pi} \tau(s)$.

Proof. The choice $B_\alpha = (1-\alpha)\varepsilon_{\text{cent},d} + \alpha\varepsilon_{A,\text{cand},d}$ upper-bounds Ξ_α . Substitute it in Eq. (27). $\square$

I.1 SAFE-STEP BOUNDARY CASES

For Corollary N.3, the zero step $\alpha = 0$ is equality. At $\alpha = 1$, the bound certifies the full candidate step exactly when $M_d \geq \gamma\Delta_g$. The remaining nonredundant cases are:

- if $\gamma = 0$ and $M_d \geq 0$, every $\alpha \in [0, 1]$ is certified;
- if $M_d < 0$, no positive step is certified by this bound;
- if $\gamma > 0$, $M_d = 0$, and $\Delta_g > 0$, no positive step is certified;
- if $M_d = \Delta_g = 0$, every step has a zero lower bound;
- if $\gamma > 0$, $M_d > 0$, and $M_d \geq \gamma\Delta_g$, every $\alpha \in [0, 1]$ is certified;
- if $\gamma > 0$ and $0 < M_d < \gamma\Delta_g$, every

$$\alpha \leq \frac{M_d(1-\gamma)}{\gamma(\Delta_g - M_d)} \quad (130)$$

is certified. The denominator is positive and the threshold lies strictly between zero and one.

Proof of Corollary N.3. The invariance of $\mathcal{C}_{\text{pair}}$ under both endpoint policies implies its invariance under every π_α . Hence Theorem 7.3 applies for each $\alpha \in [0, 1]$ with the same choice $\mathcal{C}_{\text{mix}} = \mathcal{C}_{\text{pair}}$, the same Δ_g , estimator, defect function, and one-sided defect bound. Exact centering removes the current-policy component:

$$\mathbb{E}_{s \sim d_\pi, a \sim \pi_\alpha(\cdot|s)} \widehat{A}(s, a) = \alpha\widehat{g}.$$

Together with the definition of the fixed candidate defect b , this gives the candidate-slope identity required by the certificate,

$$\widehat{g} = \mathbb{E}_{s \sim d_\pi} [g(s) + b(s)]. \quad (131)$$

Consequently,

$$\mathbb{E}_{d_\pi} g = \widehat{g} - \mathbb{E}_{d_\pi} b \geq \widehat{g} - \bar{\beta}_d = M_d. \quad (132)$$

The performance-difference identity, the span-TV expectation inequality, and Lemma 7.2 now give

$$\eta(\pi_\alpha) - \eta(\pi) = \frac{\alpha}{1-\gamma} \mathbb{E}_{d_\pi} g \geq \frac{\alpha}{1-\gamma} \left[\mathbb{E}_{d_\pi} g - \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha} \right] \geq \frac{\alpha}{1-\gamma} \left[M_d - \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha} \right].$$

This proves Eqs. (157) and (158). The exact signed choice meets Eq. (155) with equality, while $\mathbb{E}b \leq \mathbb{E}|b| \leq \varepsilon_{A,\text{cand}}$ proves the other two specializations. For

$$\varphi(\alpha) := \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha}, \quad \varphi'(\alpha) = \frac{\gamma\Delta_g(1-\gamma)}{(1-\gamma+\gamma\alpha)^2} \geq 0.$$

Thus $\varphi(0) = 0$ and $\max_{\alpha \in [0,1]} \varphi(\alpha) = \varphi(1) = \gamma\Delta_g$. If $\gamma > 0$ and $0 < M_d < \gamma\Delta_g$, then $M_d < \Delta_g$ because $\gamma < 1$, so $\Delta_g - M_d > 0$ and the threshold is positive. Moreover,

$$\frac{M_d(1-\gamma)}{\gamma(\Delta_g - M_d)} < 1 \iff M_d < \gamma\Delta_g,$$

which is exactly the branch premise. Rearrangement therefore gives Eq. (130) directly, with no outer minimum or further intersection with $[0, 1]$. The monotonicity and direct substitution give every stated boundary case. In particular, $\varphi(1) = \gamma\Delta_g$ places $M_d = \gamma\Delta_g$ in the all-step branch, and when $M_d = \Delta_g = 0$ the displayed lower bound is zero for every step. $\square$

I.2 RESIDUAL-TO-CPI SUBSTITUTIONS AND PROOF

Corollary I.2 (Self-bootstrap and target-network residual certificates). *Use the measurable Bellman domain of Theorem 5.1 with $\mathcal{C} = \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$. Take $U_n \in B_b(\mathcal{C})$ and use the common scalar boundary $U_n(s_{\text{abs}}) = V^\pi(s_{\text{abs}}) = b$. Define*

$$\varepsilon_{\text{res},n} \triangleq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (133)$$

Then

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{\varepsilon_{\text{res},n}}{1 - \gamma^K}. \quad (134)$$

If $\hat{A} = \hat{A}_{U_n}$ is the exact one-step centered estimator of Eq. (145), with its transition expectation and baseline sum under π evaluated exactly, then Assumption 7.1 holds with

$$\varepsilon_A \triangleq \frac{2\gamma \varepsilon_{\text{res},n}}{1 - \gamma^K}. \quad (135)$$

Consequently, under Theorem 7.3,

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{1-\gamma} \frac{\varepsilon_{\text{res},n}}{1-\gamma^K} - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (136)$$

For a finite depth $q \geq 0$, instead suppose Proposition 6.1 holds on the same domain with $\mathcal{E} = \mathcal{C}$. Then

$$\|U_q - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{R_{\text{targ},q}^P}{1 - \gamma^K}. \quad (137)$$

For finite depths $0 \leq n < m$, if $U_n, U_m \in B_b(\mathcal{C})$ use the same scalar boundary and the depth- m target-network quantities are established, then

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{R_{\text{targ},m}^P}{1 - \gamma^K}. \quad (138)$$

For the depth- q target statement, take $\hat{A} = \hat{A}_{U_q}$ from Eq. (145), with its transition expectation and current-policy baseline evaluated exactly, and define $\varepsilon_{A,\text{cand}}$ by Eq. (30). Then

$$\varepsilon_{A,\text{cand}} \leq 2\gamma\tau_\infty \frac{R_{\text{targ},q}^P}{1 - \gamma^K}. \quad (139)$$

Under the exact-mixture and scalar-span hypotheses of Theorem 7.3,

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{(1-\gamma)(1-\gamma^K)} R_{\text{targ},q}^P - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}, \quad (140)$$

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{1-\gamma} \left(\|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{R_{\text{targ},m}^P}{1 - \gamma^K} \right) - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (141)$$

The second display uses the same exact value-derived construction at depth n . Every target expression retains the propagated-lag quantity $R_{\text{targ},q}^P$ before the ordinary sup-lag fallback $R_{\text{targ},q}^{\text{sup}} \leq R_{\text{targ},q}^{\text{sup}}$.

Table 6: Centering regimes. The signed master theorem remains applicable when exact centering is unavailable.

Regime	Centering status	Theorem object
Finite enumerable actions with exact summation	Pointwise exact	$c(s) = 0$
Tractable exact integration, or deterministic quadrature with a separately certified error	Exact, or bounded numerical defect	Exact theorem, or an explicit defect bound
Sampled or learned baseline	Approximate	Signed $c(s)$, or an established B_α

Proof. Because $\mathcal{T}_K^\pi$ is a γ^K -contraction with fixed point V^π ,

$$\begin{aligned} \|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} &\leq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \|\mathcal{T}_K^\pi U_n - \mathcal{T}_K^\pi V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \\ &\leq \varepsilon_{\text{res}, n} + \gamma^K \|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \end{aligned}$$

Rearranging gives (134). Lemma J.2 yields (135), and Corollary 5.3 sharpens the candidate bias to $2\gamma\tau_\infty\varepsilon_{\text{res}, n}/(1 - \gamma^K)$. Substitution into Theorem 7.3 gives Eq. (136); $\tau_\infty \leq 1$ recovers the uniform fallback.

For the target-network forms, Proposition 6.1 and the same Bellman contraction argument give the following, where $\|\cdot\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ abbreviates $\|\cdot\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ within this proof:

$$(1 - \gamma^K) \|U_q - V^\pi\|_{\infty}^{\text{adv}} \leq \|U_q - \mathcal{T}_K^\pi U_q\|_{\infty}^{\text{adv}} \leq R_{\text{targ}, q}^P,$$

which proves Eq. (137). Adding and subtracting U_m and applying this result at $q = m$ proves Eq. (138). For the stated exact value-derived estimator, Corollary 5.3 applies with $\delta_V \leq R_{\text{targ}, q}^P/(1 - \gamma^K)$, proving Eq. (139). Substitution into the scalar specialization of Theorem 7.3 proves Eq. (140). Applying the same overlap result to the depth- n estimator with

$$\delta_V \leq \|U_n - U_m\|_{\infty}^{\text{adv}} + \frac{R_{\text{targ}, m}^P}{1 - \gamma^K}$$

proves Eq. (141). No step identifies the target-network population operator with $\mathcal{T}_K^\pi$; the former only supplies the displayed upper bound on the latter’s residual. $\square$

Remark I.3 (Self-bootstrap residual to fixed-reference residual). The uniform self-bootstrap residual below upper-bounds $\varepsilon_{\text{res}}^*$ up to the same truncation gap:

$$\varepsilon_{\text{res}}^* \leq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + (1 + \gamma^K) \|U_n - U_*\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (142)$$

This follows by adding and subtracting U_n inside $U_* - \mathcal{T}_K^\pi U_*$ and using K -step Bellman contraction on the final term. It is distinct from the target-network population residual in Proposition 6.1; Eq. (19) controls the self-bootstrap residual only after also controlling the propagated target-network lag $\|P_\pi^K(\bar{V} - U_n)\|_{\infty}$; the ordinary sup-lag $\|\bar{V} - U_n\|_{\infty}$ is the nonexpansive fallback. Under Assumption B.1 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and the Lipschitz-head condition on that domain, combining (142) with Proposition B.2 gives

$$\varepsilon_{\text{res}}^* \leq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + (1 + \gamma^K) L_V \frac{L_z^n}{1 - L_z} C_z. \quad (143)$$

Finite-batch or mean-square estimates of either residual remain diagnostics and do not establish these sup-norm inequalities.

J ADVANTAGE ESTIMATION AND BOUNDS

The value-to-advantage calculation has two parts: exact centering removes the base-policy mean, and successor value error controls what remains. The following table distinguishes exact evaluation of that mean from approximate baselines.

$$\hat{Q}_V(s, a) = \mathbb{E}_{s'}[r(s, a, s') + \gamma V(s')], \quad (144)$$

$$\hat{A}_V(s, a) = \hat{Q}_V(s, a) - \mathbb{E}_{b \sim \pi(\cdot|s)}[\hat{Q}_V(s, b)]. \quad (145)$$

Remark J.1 (Centering and baseline computation). The linear-in- α $\varepsilon_{A,\text{cand}}$ term in Theorem 7.3 relies on the *statewise centering* property

$$\mathbb{E}_{a \sim \pi(\cdot|s)}[\hat{A}(s, a)] = 0 \quad \text{for all } s.$$

This holds identically when $\hat{A}$ is defined as in (145) with the baseline computed exactly as $B_V(s) = \mathbb{E}_{b \sim \pi(\cdot|s)}[\hat{Q}_V(s, b)]$ (e.g., by summation over a finite action space).

If the baseline is approximated, centering may fail. We quantify this by the *centering defect*

$$\varepsilon_{\text{cent}} := \sup_{s \in \mathcal{C}_\pi} \left| \mathbb{E}_{a \sim \pi(\cdot|s)}[\hat{A}(s, a)] \right|.$$

The exact-centered scalar specialization in Eq. (31) corresponds to $\varepsilon_{\text{cent}} = 0$; the signed master inequality Eq. (27) also covers nonzero centering defects.

Lemma J.2 (Value to one-step advantage error). *Let $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ be the advantage-evaluation closure from Definition E.2, and let $\delta_V := \|V - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Then for every $s \in \mathcal{C}_\pi$ and for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every a ,*

$$|\hat{A}_V(s, a) - A^\pi(s, a)| \leq 2\gamma \delta_V.$$

The factor 2γ is tight worst-case because error appears in both Q and the baseline.

Remark J.3 (Extensions to K -step and GAE-style advantages). Lemma J.2 is stated for the *one-step* estimator (145), hence the factor 2γ . If instead one defines a K -step action-value estimate

$$\hat{Q}_V^{(K)}(s, a) := \mathbb{E} \left[\sum_{t=0}^{K-1} \gamma^t r_t + \gamma^K V(s_K) \mid s_0 = s, a_0 = a, \pi \right],$$

and the corresponding centered advantage $\hat{A}_V^{(K)}(s, a) := \hat{Q}_V^{(K)}(s, a) - \mathbb{E}_{b \sim \pi(\cdot|s)}[\hat{Q}_V^{(K)}(s, b)]$ (with the baseline computed exactly), then the same argument yields

$$|\hat{A}_V^{(K)}(s, a) - A^\pi(s, a)| \leq 2\gamma^K \delta_V.$$

λ -return or GAE-style estimators lie outside the displayed bridge unless their estimator-specific uniform error is separately established. Such an error may be supplied directly as ε_A in Assumption 7.1; no GAE coefficient is derived here.

Corollary J.4 (Advantage error in terms of theorem parameters). *Under Theorem 5.1 and the exact-centering conditions of Lemma J.2,*

$$\varepsilon_A \leq 2\gamma B_{n,m}. \quad (146)$$

Under Assumption B.1 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and the Lipschitz-head condition on that domain, taking $m \rightarrow \infty$ yields

$$\varepsilon_A \leq \underbrace{\frac{2\gamma \varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{architectural}} + 2\gamma L_V \underbrace{\frac{L_z^n}{1 - L_z}}_{\text{unrolling}} C_z. \quad (147)$$

Here ε_A bounds the advantage error on $\mathcal{C}_\pi$ for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every action, and the intermediate value-error quantity in Lemma J.2 is taken over the advantage-evaluation closure $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ (Definition E.2).

K PROOFS

K.1 LEMMA K.1 (INTERIOR PROJECTION FIXED POINTS)

Lemma K.1 (Interior projection fixed points). *Let $\Pi_R : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be the Euclidean projection onto the closed ball $\{z : \|z\|_2 \leq R\}$ as in (34). Suppose $z^* = \Pi_R(u)$ for some $u \in \mathbb{R}^d$. If $\|z^*\|_2 < R$, then $z^* = u$ (i.e., the projection is inactive at u). Consequently, if z^* is a fixed point of $f_\theta(z, y, x) = \Pi_R(f_\theta(z, y, x))$ and $\|z^*\|_2 < R$, then $f_\theta(z^*, y, x) = z^*$ and Π_R is inactive at $f_\theta(z^*, y, x)$.*

Proof. By (34), if $\|u\|_2 \leq R$ then $\Pi_R(u) = u$. Since $z^* = \Pi_R(u)$ lies in the open ball $\|z^*\|_2 < R$, we have $z^* \neq R \frac{u}{\|u\|_2}$, so the second case of (34) cannot apply; hence $\|u\|_2 \leq R$ and $\Pi_R(u) = u = z^*$. For the consequence: if $z^* = \tilde{f}_\theta(z^*, y, x) = \Pi_R(f_\theta(z^*, y, x))$ and $\|z^*\|_2 < R$, applying the first statement with $u = f_\theta(z^*, y, x)$ gives $f_\theta(z^*, y, x) = z^*$. $\square$

K.2 PROOF OF PROPOSITION B.2

Proof. Fix $s \in \mathcal{D}^\circ$. By Assumption B.1, T_s maps $\mathcal{Z}_{\text{inv}}$ to itself (forward invariance) and is a contraction with modulus $L_z < 1$. Since $\mathcal{Z}_{\text{inv}}$ is a closed subset of the complete metric space $(\mathcal{Z}, \|\cdot\|)$, it is itself complete.

Banach's theorem yields a unique fixed point $z^*(s) \in \mathcal{Z}_{\text{inv}}$ and linear convergence (37).

Under the additional measurability premises in Eq. (39), the iterates are measurable by induction. Indeed, if $z^{(j)}$ is measurable, then $s \mapsto (s, z^{(j)}(s))$ is measurable into the product space, so $z^{(j+1)}(s) = T(s, z^{(j)}(s))$ is measurable. Banach convergence gives $z^{(j)}(s) \rightarrow z^*(s)$ pointwise. For every nonempty closed $F \subseteq \mathcal{Z}_{\text{inv}}$, continuity of distance and pointwise convergence give

$$d(z^*(s), F) = \lim_{j \rightarrow \infty} d(z^{(j)}(s), F).$$

The right side is a pointwise limit of measurable real functions, and

$$\{s : z^*(s) \in F\} = \{s : d(z^*(s), F) = 0\}.$$

The empty closed set has empty preimage. Since closed sets generate the Borel σ -algebra, z^* is measurable.

For the finite-unrolling bound, specialize to $\mathcal{D}^\circ = \mathcal{C}^\circ$ and take $z^{(0)}(s) = z_{\text{init}}(x(s), y(s))$:

$$z^*(s) - z^{(n)}(s) = \sum_{k=n}^{\infty} (z^{(k+1)}(s) - z^{(k)}(s)).$$

Contraction gives $\|z^{(k+1)}(s) - z^{(k)}(s)\| \leq L_z^k \|z^{(1)}(s) - z^{(0)}(s)\|$, so

$$\|z^*(s) - z^{(n)}(s)\| \leq \|z^{(1)}(s) - z^{(0)}(s)\| \sum_{k=n}^{\infty} L_z^k = \frac{L_z^n}{1 - L_z} \|z^{(1)}(s) - z^{(0)}(s)\|.$$

By definition (36), using the declared update $\tilde{f}_\theta$,

$$\|z^{(1)}(s) - z^{(0)}(s)\| = \|T_s(z_{\text{init}}(x(s), y(s))) - z_{\text{init}}(x(s), y(s))\| \leq C_z.$$

□

K.3 PROOF OF LEMMA C.9

Proof. For $T_{x,y_t}^{\circ n}$ a κ_n -contraction (where $\kappa_n = L_z^n$):

$$\|z_{t+1} - z^*(x, y_t)\| = \|T_{x,y_t}^{\circ n}(z_t) - T_{x,y_t}^{\circ n}(z^*(x, y_t))\| \leq \kappa_n e_t.$$

By Assumption C.5:

$$\|z^*(x, y_t) - z^*(x, y_{t+1})\| \leq L_{z^*} \|y_{t+1} - y_t\|_{\mathcal{E}} \leq L_{z^*} \Delta y_{\max}.$$

Thus:

$$e_{t+1} = \|z_{t+1} - z^*(x, y_{t+1})\| \leq \|z_{t+1} - z^*(x, y_t)\| + \|z^*(x, y_t) - z^*(x, y_{t+1})\| \leq \kappa_n e_t + L_{z^*} \Delta y_{\max}.$$

Unrolling this recurrence:

$$e_t \leq \kappa_n^t e_0 + L_{z^*} \Delta y_{\max} \sum_{j=0}^{t-1} \kappa_n^j = \kappa_n^t e_0 + \frac{L_{z^*} \Delta y_{\max} (1 - \kappa_n^t)}{1 - \kappa_n}.$$

Using $e_0 \leq E_0$ and noting that this expression is maximized either at $t = 0$ (giving E_0) or as $t \rightarrow \infty$ (giving $\frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n}$), we have:

$$\sup_t e_t \leq \max \left(E_0, \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n} \right) \leq E_0 + \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n}.$$

For the used latent:

$$\|z_t^{(n)} - z^*(x, y_t)\| = \|T_{x, y_t}^{\circ n}(z_t) - z^*(x, y_t)\| = \|T_{x, y_t}^{\circ n}(z_t) - T_{x, y_t}^{\circ n}(z^*(x, y_t))\| \leq \kappa_n e_t,$$

where the second equality uses that $z^*(x, y_t)$ is the fixed point of T_{x, y_t} , hence also of $T_{x, y_t}^{\circ n}$, and the inequality follows from κ_n -contractivity of the n -fold composition. Thus:

$$\sup_t \|z_t^{(n)} - z^*(x, y_t)\| \leq \kappa_n \sup_t e_t \leq \kappa_n \max\left(E_0, \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n}\right) = C_{\text{drift}}(n),$$

and the final “ $\leq$ ” in (73) gives the simpler additive upper bound. $\square$

K.4 PROOF OF THEOREM 5.1

Proof. Write $\|\cdot\|_\infty$ for the ordinary sup-norm on $\mathcal{C}$. Invariance and the Markov-kernel measurability property imply $P_\pi^K V \in B_b(\mathcal{C})$ whenever $V \in B_b(\mathcal{C})$; bounded r_K^π therefore makes $\mathcal{T}_K^\pi$ a self-map. For $V, W \in B_b(\mathcal{C})$,

$$\|\mathcal{T}_K^\pi V - \mathcal{T}_K^\pi W\|_\infty \leq \gamma^K \|V - W\|_\infty.$$

The bounded-reward premise makes the ordinary discounted policy value well defined, bounded, and measurable, with the uniformly convergent representation

$$V^\pi = \sum_{t=0}^{\infty} \gamma^t P_\pi^t r^\pi.$$

Splitting this series after K terms gives $V^\pi = \mathcal{T}_K^\pi V^\pi$. Completeness and Banach’s theorem then identify this policy value as the unique fixed point of $\mathcal{T}_K^\pi$. Thus

$$\begin{aligned} \|U_m - V^\pi\|_\infty &\leq \|U_m - \mathcal{T}_K^\pi U_m\|_\infty + \|\mathcal{T}_K^\pi U_m - \mathcal{T}_K^\pi V^\pi\|_\infty \\ &\leq \|U_m - \mathcal{T}_K^\pi U_m\|_\infty + \gamma^K \|U_m - V^\pi\|_\infty. \end{aligned}$$

Rearranging and then applying the triangle inequality to $U_n - U_m + U_m - V^\pi$ proves (4).

For every nonabsorbing s , Lipschitzness of the scalar head and the telescoping latent path give

$$\begin{aligned} |U_n(s) - U_m(s)| &\leq L_V \|z^{(n)}(s) - z^{(m)}(s)\| \\ &\leq L_V \sum_{j=n}^{m-1} \|z^{(j+1)}(s) - z^{(j)}(s)\|. \end{aligned}$$

The left side is zero at s_{abs} because its value is depth independent. Taking the supremum only over $\mathcal{C}^\circ$ for the latent endpoints proves Eq. (5). The triangle inequality for the telescoping latent path proves Eq. (6); using $\sup_s \sum_j g_j(s) \leq \sum_j \sup_s g_j(s)$ proves Eq. (7). No absorbing latent is defined.

Under the $\mathcal{D}^\circ = \mathcal{C}^\circ$ invocation of Assumption B.1,

$$\|z^{(j+1)}(s) - z^{(j)}(s)\| \leq L_z^j \|z^{(1)}(s) - z^{(0)}(s)\| \leq L_z^j C_z.$$

Summing from $j = n$ through $m - 1$ yields (43). Proposition B.2 and the Lipschitz head give $\|U_m - U_*\|_\infty \rightarrow 0$. Moreover,

$$|\|U_m - \mathcal{T}_K^\pi U_m\|_\infty - \|U_* - \mathcal{T}_K^\pi U_*\|_\infty| \leq (1 + \gamma^K) \|U_m - U_*\|_\infty \rightarrow 0.$$

Taking $m \rightarrow \infty$ therefore proves (44). $\square$

Remark K.2 (Finite- R specialization of the truncation constant). Under the Appendix L initialization condition $\|z_{\text{init}}(x(s), y(s))\|_2 \leq R$ for every $s \in \mathcal{C}^\circ$, with $\mathcal{Z}_{\text{inv}} = \{z : \|z\|_2 \leq R\}$, the projection Π_R ensures $\|T_s(z_{\text{init}}(x(s), y(s)))\|_2 \leq R$ on the same domain. Therefore

$$C_z = \sup_{s \in \mathcal{C}^\circ} \|\tilde{f}_\theta(z_{\text{init}}(x(s), y(s)), y(s), x(s)) - z_{\text{init}}(x(s), y(s))\|_2 \leq 2R.$$

The contractive specialization of Theorem 5.1 therefore gives

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{\varepsilon_{\text{res}}^*}{1 - \gamma^K} + 2L_V \frac{L_z^n}{1 - L_z} R.$$

This makes R explicit only through the finite-unrolling term; it does not identify the dependence of $\varepsilon_{\text{res}}^*$ on R .

K.5 PROJECTION-ACTIVE THEORY

Projection-aware CPI penalty (full statement). Assume all premises of Proposition K.3 on $\mathcal{C}^\circ$, including saturation and the represented-initialization condition, together with the zero-anchor hypotheses of Proposition K.4, and suppose $\bar{L}_z^{\text{sat}}(R) < 1$. That proposition establishes Assumption B.1 on $\mathcal{D}^\circ = \mathcal{C}^\circ$. Combining Eqs. (45) and (46) with $C_z \leq 2R$ yields a projection-aware version of (47):

$$\eta(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma\tau_\infty}{1-\gamma} \left(\frac{\varepsilon_0 + (1+\gamma^K)L_V(R)R}{1-\gamma^K} + 2L_V(R) \frac{(\bar{L}_z^{\text{sat}}(R))^n}{1-\bar{L}_z^{\text{sat}}(R)} R \right) - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (148)$$

Projection affects both the architectural and truncation terms. The displayed bound does not make R a monotone performance dial because the residual, head Lipschitz factor, saturation margin, and optimizer trajectory may all change with the projection regime.

Proposition K.3 (Projection-induced contraction on a saturated annulus). *Fix $R > 0$, let $B_R := \{z : \|z\|_2 \leq R\}$, and, for $s \in \mathcal{C}^\circ$, set $T_s(z) := \Pi_R(f_\theta(z, y(s), x(s)))$. Define the pre-projection domain-relative Lipschitz constant*

$$L_z^{\text{pre}}(R) := \sup_{s \in \mathcal{C}^\circ} \sup_{\substack{z, z' \in B_R \\ z \neq z'}} \frac{\|f_\theta(z, y(s), x(s)) - f_\theta(z', y(s), x(s))\|_2}{\|z - z'\|_2}.$$

Assume there exists $\rho_R > R$ such that

$$\|f_\theta(z, y(s), x(s))\|_2 \geq \rho_R \quad \text{for all } s \in \mathcal{C}^\circ \text{ and } z \in B_R.$$

Then, for every $s \in \mathcal{C}^\circ$ and $z, z' \in B_R$,

$$\|T_s(z) - T_s(z')\|_2 \leq \bar{L}_z^{\text{sat}}(R) \|z - z'\|_2, \quad \bar{L}_z^{\text{sat}}(R) := \frac{R}{\rho_R} L_z^{\text{pre}}(R).$$

Projection also gives $T_s(B_R) \subseteq B_R$. Hence, if $\bar{L}_z^{\text{sat}}(R) < 1$ and $z_{\text{init}}(x(s), y(s)) \in B_R$ for every $s \in \mathcal{C}^\circ$, then Assumption B.1 holds with $\mathcal{D}^\circ = \mathcal{C}^\circ$, $\mathcal{Z}_{\text{inv}} = B_R$, and $L_z \leq \bar{L}_z^{\text{sat}}(R)$. This conclusion does not assert contraction for input pairs not represented in $\mathcal{C}^\circ$.

Proof. For any u, v with $\|u\|_2, \|v\|_2 \geq \rho_R > R$,

$$\|\Pi_R(u) - \Pi_R(v)\|_2 = R \left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2.$$

Using

$$\left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2^2 = \frac{\|u - v\|_2^2 - (\|u\|_2 - \|v\|_2)^2}{\|u\|_2 \|v\|_2} \leq \frac{\|u - v\|_2^2}{\rho_R^2},$$

we obtain

$$\|\Pi_R(u) - \Pi_R(v)\|_2 \leq \frac{R}{\rho_R} \|u - v\|_2.$$

Fix $s \in \mathcal{C}^\circ$ and apply this with $u = f_\theta(z, y(s), x(s))$ and $v = f_\theta(z', y(s), x(s))$ for $z, z' \in B_R$. Then

$$\|T_s(z) - T_s(z')\|_2 \leq \frac{R}{\rho_R} L_z^{\text{pre}}(R) \|z - z'\|_2. \quad \square$$

Proposition K.4 (Projection-aware residual via a state-dependent latent anchor). *Fix the discounted MDP, policy pair, γ , K , closure $\mathcal{C}$, and common absorber from Corollary I.2. Require its bounded-function hypotheses, including $\mathcal{T}_K^\pi : B_b(\mathcal{C}) \rightarrow B_b(\mathcal{C})$. Write $\mathcal{C}^\circ = \mathcal{C} \setminus \{s_{\text{abs}}\}$. Assume Assumption B.1 holds on $\mathcal{D}^\circ = \mathcal{C}^\circ$ with $\mathcal{Z}_{\text{inv}} = B_R$. Assume $(s, z) \mapsto T_s(z)$ is jointly measurable on $\mathcal{C}^\circ \times B_R$ and $s \mapsto z_{\text{init}}(x(s), y(s))$ is measurable into B_R . Proposition B.2 then makes the represented fixed-point map z^* measurable. Invoke Assumption B.3 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and $\mathcal{Z}_{\text{bd}} = B_R$,*

and require that the induced evaluator $U_*(s) = V_\psi(z^*(s), x(s), y(s))$ belongs to $B_b(\mathcal{C})$ with the common absorbing boundary. Fix any measurable anchor $z_{\text{anc}} : \mathcal{C}^\circ \rightarrow B_R$ and define

$$\begin{aligned} B_{\text{anc}}(s) &:= V_\psi(z_{\text{anc}}(s), x(s), y(s)), & s \in \mathcal{C}^\circ, \\ B_{\text{anc}}(s_{\text{abs}}) &:= -C_{\text{max}}, \\ \varepsilon_{\text{anc}} &:= \|B_{\text{anc}} - \mathcal{T}_K^\pi B_{\text{anc}}\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \end{aligned}$$

Assume $B_{\text{anc}} \in B_b(\mathcal{C})$ and let

$$\begin{aligned} D_{\text{anc}}(R) &:= \sup_{s \in \mathcal{C}^\circ} \|z^*(s) - z_{\text{anc}}(s)\|_2, \\ L_V(R) &:= \sup_{s \in \mathcal{C}^\circ} \sup_{\substack{z, z' \in B_R \\ z \neq z'}} \frac{|V_\psi(z, x(s), y(s)) - V_\psi(z', x(s), y(s))|}{\|z - z'\|_2}. \end{aligned}$$

Then

$$\varepsilon_{\text{res}}^* \leq \varepsilon_{\text{anc}} + (1 + \gamma^K) L_V(R) D_{\text{anc}}(R). \quad (149)$$

In the special case $z_{\text{anc}}(s) \equiv 0$ on $\mathcal{C}^\circ$,

$$\begin{aligned} B_0(s) &:= V_\psi(0, x(s), y(s)), & s \in \mathcal{C}^\circ, \\ B_0(s_{\text{abs}}) &:= -C_{\text{max}}, \\ \varepsilon_{\text{res}}^* &\leq \varepsilon_0 + (1 + \gamma^K) L_V(R) R, \\ \varepsilon_0 &:= \|B_0 - \mathcal{T}_K^\pi B_0\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \end{aligned}$$

Proof. Add and subtract B_{anc} :

$$U_* - \mathcal{T}_K^\pi U_* = (B_{\text{anc}} - \mathcal{T}_K^\pi B_{\text{anc}}) + (U_* - B_{\text{anc}}) + (\mathcal{T}_K^\pi B_{\text{anc}} - \mathcal{T}_K^\pi U_*).$$

Taking $\|\cdot\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ and using the γ^K -contraction of $\mathcal{T}_K^\pi$ on the difference of its two images gives

$$\varepsilon_{\text{res}}^* \leq \varepsilon_{\text{anc}} + (1 + \gamma^K) \|U_* - B_{\text{anc}}\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}.$$

For every $s \in \mathcal{C}^\circ$, both $z^*(s)$ and $z_{\text{anc}}(s)$ lie in B_R , so

$$\begin{aligned} |U_*(s) - B_{\text{anc}}(s)| &= |V_\psi(z^*(s), x(s), y(s)) - V_\psi(z_{\text{anc}}(s), x(s), y(s))| \\ &\leq L_V(R) \|z^*(s) - z_{\text{anc}}(s)\|_2 \leq L_V(R) D_{\text{anc}}(R). \end{aligned}$$

At the absorbing boundary, $U_*(s_{\text{abs}}) = B_{\text{anc}}(s_{\text{abs}}) = -C_{\text{max}}$, so the difference is zero. The zero-anchor specialization uses $D_{\text{anc}}(R) \leq R$. $\square$

K.6 PROOF OF LEMMA J.2

Proof. For any (s, a) ,

$$\widehat{Q}_V(s, a) - Q^\pi(s, a) = \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V(s') - V^\pi(s')].$$

If $s \in \mathcal{C}_\pi$, Definition E.2 gives $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} | s, a) = 1$ for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every a . Thus $|V(s') - V^\pi(s')| \leq \delta_V$ for transition-almost every s' , and hence $|\widehat{Q}_V(s, a) - Q^\pi(s, a)| \leq \gamma \delta_V$.

Similarly,

$$|\mathbb{E}_{b \sim \pi(\cdot | s)} \widehat{Q}_V(s, b) - \mathbb{E}_{b \sim \pi(\cdot | s)} Q^\pi(s, b)| \leq \gamma \delta_V,$$

since $b \sim \pi(\cdot | s)$ and the same almost-sure closure condition applies. Subtracting these two bounds yields $|\widehat{A}_V(s, a) - A^\pi(s, a)| \leq 2\gamma \delta_V$. $\square$

K.7 PROOF OF COROLLARY C.2

Proof. The clock-complete state, frozen shared map F_n , and specified initial law define one augmented MDP for both action kernels. On its bounded measurable functions, $\bar{T}_K^\pi$ is a γ^K -contraction in the augmented closure norm. In this proof, $\|\cdot\|_\infty^{\text{adv}}$ abbreviates $\|\cdot\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}$. Thus

$$\begin{aligned} \|\tilde{U}_n - V^\pi\|_\infty^{\text{adv}} &\leq \|\tilde{U}_n - \bar{T}_K^\pi \tilde{U}_n\|_\infty^{\text{adv}} + \|\bar{T}_K^\pi \tilde{U}_n - \bar{T}_K^\pi V^\pi\|_\infty^{\text{adv}} \\ &\leq \bar{\varepsilon}_{\text{res}, n} + \gamma^K \|\tilde{U}_n - V^\pi\|_\infty^{\text{adv}}. \end{aligned}$$

Rearrangement proves (55). Lemma J.2, applied on the augmented closure with exact centering conditional on $\bar{s}$, gives (56). Corollary 5.3 supplies the $\bar{\tau}_\infty$ refinement. Under the exact pointwise mixture, substitution into Theorem 7.3 proves (59).

For the target-network statements, add and subtract its population backup on the same fixed augmented MDP:

$$\tilde{U}_q - \bar{T}_K^\pi \tilde{U}_q = \left[\tilde{U}_q - (\bar{r}_K^\pi + \gamma^K P_\pi^K \bar{V}) \right] + \gamma^K P_\pi^K (\bar{V} - \tilde{U}_q).$$

The triangle inequality and nonexpansiveness of P_π^K prove Eq. (63). Bellman contraction then gives

$$(1 - \gamma^K) \|\tilde{U}_q - V^\pi\|_\infty^{\text{adv}} \leq \bar{R}_{\text{targ}, q}^P,$$

which proves Eq. (64). Adding and subtracting $\tilde{U}_m$ proves Eq. (65). The augmented version of Corollary 5.3, applied to the stated exact value-derived estimator, proves Eq. (67). Substitution into Theorem 7.3 proves Eq. (68); applying the same argument to the depth- n estimator and Eq. (65) proves Eq. (69). Throughout, q changes only an endpoint function, not F_n , the policies, or the augmented kernel. $\square$

For completeness, the optional slow-drift comparisons follow directly. Lemma C.9 and the Lipschitz head imply (75); adding and subtracting $\tilde{U}_*$ and applying the residual bound (1) proves (76). Finally, the reverse triangle inequality and Bellman contraction give

$$\begin{aligned} |\bar{\varepsilon}_{\text{res}, n} - \bar{\varepsilon}_{\text{res}, *}| &\leq \|(\tilde{U}_n - \tilde{U}_*) - (\bar{T}_K^\pi \tilde{U}_n - \bar{T}_K^\pi \tilde{U}_*)\|_\infty^{\text{adv}} \\ &\leq (1 + \gamma^K) \|\tilde{U}_n - \tilde{U}_*\|_\infty^{\text{adv}}, \end{aligned}$$

which proves (77).

L IMPLEMENTATION DETAILS

L.1 ALGORITHM PSEUDOCODE

Algorithm 2 presents one fixed-base UPI-TRM proposal in persistent- z form with an explicit remaining-budget clock. The analyzed object is one proposal at one frozen parameter snapshot; it is not a recursively promoted policy sequence. For the episodic- z variant, use $s_k = (x, y_k, h_k)$, omit the carried latent from replay, and reinitialize $z_k \leftarrow z_{\text{init}}(x, y_k)$ at the start of each loop iteration. For a collected segment of length $\ell \leq K$, let d_{seg} indicate that its final transition terminates. The folded-reward implementation target is

$$G_{\text{fold}} = \sum_{k=0}^{\ell-1} \gamma^k r_k + \mathbf{1}\{-d_{\text{seg}}\} \gamma^\ell \bar{V}(\bar{s}_\ell). \quad (150)$$

Here $d_{\text{seg}} = 0$ implies $\ell = K$; early nonterminal segments are not admissible exact- K targets. On a terminal segment, the stored final reward is the formal terminal-transition reward plus γb , where $b = V^\pi(s_{\text{abs}})$. The zero padding and zero terminal bootstrap in Eq. (150) are therefore an accounting representation of the formal absorbing tail, not an additional self-loop. Under frozen-current-policy collection, this target has conditional population backup $r_K^\pi + \gamma^K P_\pi^K \bar{V}$ in the episodic model. Hence Proposition 6.1 converts its uniform population residual to the theorem's self-bootstrap residual after adding the propagated lag $\gamma^K \|P_\pi^K (\bar{V} - U_n)\|_\infty$. Nonexpansiveness gives the ordinary sup-lag

fallback $\gamma^K \|\bar{V} - U_n\|_\infty$. In the persistent model, replace every Bellman object by its augmented-MDP counterpart: in particular, $(\mathcal{E}, P_\pi, U_n, \mathcal{T}_K^\pi)$ becomes $(\bar{\mathcal{C}}, P_\pi, \bar{U}_n, \bar{\mathcal{T}}_K^\pi)$, and the expected-reward term is the augmented K -step expected reward. The sharper lag term is then $\gamma^K \|P_\pi^K(\bar{V} - \bar{U}_n)\|_\infty$, before the fallback $\gamma^K \|\bar{V} - \bar{U}_n\|_\infty$. Empirical regression loss, population L^2 error, and a finite-batch maximum establish neither required sup norm.

Algorithm 2 UPI-TRM: one fixed-base K -step collection and proposal

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1: Input:  $x \in \mathcal{X}$ ,  $y_0 \in \mathcal{Y}$ , and  $h_0 \in \mathbb{N}_{\geq 1}$ ; for persistent- $z$  only,  $z_0 \in \mathcal{Z}$ 
2:  $\gamma \in [0, 1)$ ,  $n \in \mathbb{N}_0$ ,  $K \in \mathbb{N}_{\geq 1}$ ,  $\alpha \in [0, 1]$ , and target-retention coefficient  $\tau \in [0, 1]$ 
3: Recurrent mode: either  $R > 0$  with  $P_{\text{rec}} = \Pi_R$ , or projection disabled with  $P_{\text{rec}} = \text{Id}_{\mathcal{Z}}$ ; set  $\tilde{f}_\theta = P_{\text{rec}} \circ f_\theta$ 
4: Initialize  $\bar{s}_0 \leftarrow (x, y_0, z_0, h_0)$  and collected length  $\ell \leftarrow 0$ 
5: for  $k = 0$  to  $K - 1$  do
6:    $z_k^{(n)} \leftarrow \tilde{f}_\theta^{(n)}(z_k, y_k, x)$  //  $n$  inner steps
7:   Sample  $a_k \sim \pi_\phi(a \mid y_k, z_k^{(n)}, x)$ 
8:    $y_{k+1} \leftarrow \text{edit}(y_k, a_k; x)$ ; receive  $r_k$  and terminal flag  $d_k$ ; set  $h_{k+1} \leftarrow h_k - 1$ 
9:   if  $d_k$  or  $h_{k+1} = 0$  then
10:     $\bar{s}_{k+1} \leftarrow s_{\text{abs}}$ ; store  $(\bar{s}_k, a_k, r_k, s_{\text{abs}})$  in replay
11:     $\ell \leftarrow k + 1$ ; break
12:   else
13:     $z_{k+1} \leftarrow z_k^{(n)}$  // Carry only after a nonterminal transition
14:     $\bar{s}_{k+1} \leftarrow (x, y_{k+1}, z_{k+1}, h_{k+1})$ 
15:    Store  $(\bar{s}_k, a_k, r_k, \bar{s}_{k+1})$  in replay;  $\ell \leftarrow k + 1$ 
16:   end if
17: end for
18: Compute  $G_{\text{fold}}$  for the collected length  $\ell$  via (150)
19: Fit the value head  $\psi \rightarrow \psi^+$ ; the fixed-base protocol keeps the recurrent map  $\theta$  frozen
20: Form advantages  $\hat{A}$  via centered estimator (see below)
21: Policy-head step with  $F_n$  frozen  $\rightarrow$  candidate  $\pi_{\text{cand}}$ 
22:  $\pi_{\text{new}} \leftarrow (1 - \alpha)\pi_\phi + \alpha\pi_{\text{cand}}$ 
23: Deploy the exact pointwise mixture for the theorem; optional distillation or parameter interpolation is outside it
24:  $\bar{\psi}^+ \leftarrow \tau\bar{\psi} + (1 - \tau)\psi^+$  //  $\tau$  retains the old target
25: Output: fitted head  $\psi^+$ , candidate  $\pi_{\text{cand}}$ , exact mixture  $\pi_{\text{new}}$ , and updated target head  $\bar{\psi}^+$ 

```

The fixed-base implementation seals the base policy and recurrent map F_n before collection. Replay comes from that base; optimization changes the policy-independent value head and one candidate policy head. Repeated candidate optimizer steps remain internal to the same proposal. Its exact mixture is evaluated or deployed without replacing the base.

The reference implementation stores the carried input and successor latents, the remaining-edit clock, terminal cause, terminal mask, and behavior log probability in replay. Recomputed episodic latents use the current evaluator, while persistent updates consume the recorded augmented state. On a terminal record, the environment has already folded the common absorbing tail into the reward, and every target path replaces the successor-value contribution by zero with a nonfinite-safe selection. The serialized terminal successor tensors are therefore never bootstrapped. This folded representation is return-equivalent to the single s_{abs} used in the proofs; it does not define puzzle-specific absorbing states.

Advantage estimation. The theory uses the centered estimator (145). The $\alpha\varepsilon_{A,\text{cand}}/(1 - \gamma)$ term in the readable specialization (31) requires *exact* statewise centering $\mathbb{E}_{a \sim \pi}[\hat{A}(s, a)] = 0$. This holds whenever the estimator is constructed to be exactly centered. For Eq. (145), the baseline $B_V(s) = \mathbb{E}_{b \sim \pi(\cdot|s)}[\hat{Q}_V(s, b)]$ must be evaluated exactly; the finite discrete implementation uses explicit summation. With a learned value baseline, as in standard GAE, centering is only approximate and must enter through the signed defect $c(s)$ or an upper bound B_α in Theorem 7.3. GAE is not covered by the displayed advantage bridge unless its uniform estimator error is supplied separately.

Operator-norm control and contraction-oriented details. The implementation can optionally apply operator-norm clamping on the inner $z \rightarrow z$ recursion (periodic rescaling using power-iteration estimates), followed by per-layer output scaling or an explicit scalar factor. The fixed-base exact path

rejects scheduled clamping that would mutate the frozen recurrent map. These interventions do not certify the uniform modulus on the declared domain in Assumption B.1. A possible local mini-batch diagnostic is the finite-difference ratio

$$\hat{L}_z = \max_i \frac{\|f_\theta(z_i, y_i, x_i) - f_\theta(z'_i, y_i, x_i)\|}{\|z_i - z'_i\|}.$$

Enforcing forward-invariance. For an invocation on $\mathcal{D}^\circ$, the theorem needs $z_{\text{init}}(x(s), y(s)) \in \mathcal{Z}_{\text{inv}}$ and forward invariance only for represented $s \in \mathcal{D}^\circ$. A global projection-enabled implementation is one sufficient way to enforce these conditions. In projection-disabled mode, $P_{\text{rec}} = \text{Id}_{\mathcal{Z}}$ and no projection-induced invariant ball follows; R is absent rather than set to zero.

- With $R > 0$, use an initialization $z_{\text{init}}(x, y)$ with $\|z_{\text{init}}(x, y)\|_2 \leq R$, $\mathcal{Z}_{\text{inv}} = \{z : \|z\|_2 \leq R\}$. In the reference implementation, the initial latent comes from learned latent seeds (optionally augmented by an initialization encoder) and is projected into the same ball.
- After each update, apply Euclidean projection onto the ball: $z \leftarrow \Pi_R(z)$ with Π_R as in (34).
- Under the Euclidean norm this projection is 1-Lipschitz (non-expansive) and therefore cannot increase an independently established contraction modulus.
- Equivalently, define $\tilde{f}_\theta := \Pi_R \circ f_\theta$ and unroll $\tilde{f}_\theta^n$: $z^{(t+1)} = \Pi_R(f_\theta(z^{(t)}, y, x))$ for $t = 0, \dots, n-1$.

For persistent- z , one possible diagnostic is the fresh-initialization discrepancy $\hat{C}_{\text{fresh}}(n) = \max_t \|z_t^{(n)} - z_{\text{init}}(x, y_t)\|$. It measures separation from a fresh initialization; it is not a fixed-point tracking bound and is not a substitute for $C_{\text{drift}}(n)$ or the augmented residual.

Distillation. When distilling π_{new} to π_ϕ , we minimize:

$$\mathcal{L}_{\text{distill}}(\phi) = \mathbb{E}_{\bar{s} \sim \text{buffer}} [\text{KL}(\pi_{\text{new}}(\cdot | \bar{s}) \| \pi_\phi(\cdot | \bar{s}))].$$

This buffer-averaged objective is a heuristic diagnostic, not a uniform discrepancy certificate. The distilled policy is not the exact mixture. Corollary H.8 covers it only when a uniform TV or KL discrepancy is established on the deployment domain; a finite-batch $\text{KL}(\pi_{\text{new}} \| \pi_\phi)$ does not establish that premise. Parameter interpolation is also a different policy and must not be assigned a zero realization gap from the exact pointwise mixture.

M MAPPING TO ORIGINAL TRM ARCHITECTURE

We provide an explicit mapping between our formalism and the original TRM implementation (Jolicoeur-Martineau, 2025).

Our Formalism	TRM Implementation	Notes
$z \in \mathcal{Z}$	Hidden state	Latent reasoning state
$y \in \mathcal{Y}$	Output grid/sequence	Candidate solution
f_θ	Recurrent update	Single update step
n (inner steps)	Recursion depth	Fixed or adaptively chosen evaluation budget
$z_{\text{init}}(x, y)$	Initial hidden	Learned; projected when projection is enabled
$\pi_\phi(a y, z, x)$	Task head (reinterpreted)	TRM’s task head produces the refined answer; we reframe the answer update as a discrete edit policy
$\text{edit}(y, a; x)$	Continuous refinement	TRM: continuous y -update via another forward pass; we discretize into edit actions
Episodic- z	Not in original	Our simplification
Persistent- z	Original TRM	z carried across edits
Supervised objective	Training signal	Replaced by RL reward
Adaptive halting	Early stopping	Can be related to variable n

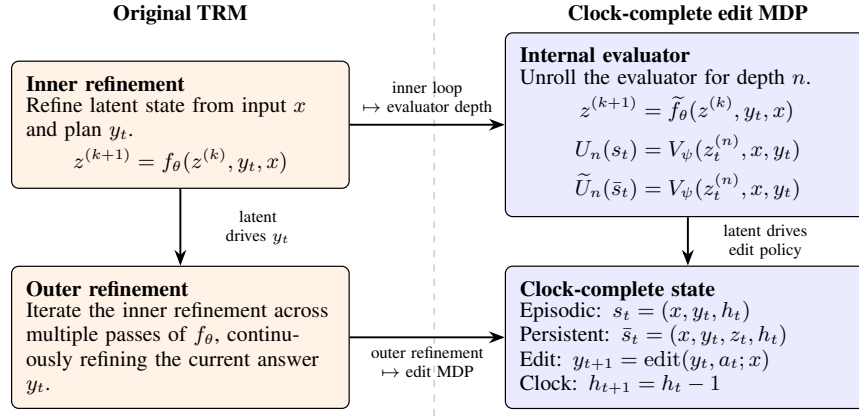


Figure 2: Internal computation and external decisions. TRM refines a latent and then its answer representation (Jolicoeur-Martineau, 2025). UPI-TRM uses the inner recurrence to evaluate a plan and replaces the outer answer update with a checker-rewarded edit. The n recurrent steps occur within one state assessment; each edit advances the environment clock. Carrying the latent between edits changes the Markov state from (x, y, h) to (x, y, z, h) .

The key differences:

1. **Training signal:** The original TRM is trained with a supervised loss; the analyzed UPI-TRM object uses checker-based rewards to construct one fixed-base candidate proposal and exact mixture.
2. **Episodic- z vs. persistent- z :** We analyze both; the original TRM is persistent- z .
3. **Halting:** The original TRM uses an adaptive halting mechanism; our analysis fixes n for tractability and treats n as an analysis parameter rather than a learned quantity.

N ADDITIONAL EXPOSITION AND PROOFS

N.1 EXTENDED RELATED WORK

Recurrent reasoning and finite computation. TRMs (Jolicoeur-Martineau, 2025) and Hierarchical Reasoning Models (Wang et al., 2025) use repeated latent updates for structured prediction; subsequent work studies training speed and test-time adaptation (Qasim & Zhang, 2025; McGovern, 2025). Deep equilibrium models instead use an equilibrium representation (Bai et al., 2019). Our primary analysis concerns a value head read after finitely many recurrent steps. The fixed-point analysis appears as a specialization when uniform contraction is established, while the finite-reference certificate also covers transient computation.

Approximate policy evaluation. Approximate dynamic programming studies how imperfect value estimates affect control (Bertsekas & Tsitsiklis, 1996; Munos, 2003; Scherrer, 2014). Munos (2003) analyzes API evaluation errors in weighted quadratic norms. The fitted-value-iteration analysis of Munos & Szepesvári (2008) includes weighted- L_p performance bounds, discounted-average concentrability, inherent Bellman residuals, and finite-sample terms. These results track errors through update sequences. Here we use an a posteriori residual bound for one policy and two finite recurrent endpoints. Our primary theorem assumes population sup-norm control; Appendix E.1 gives a weighted-norm version with an explicit fixed-kernel occupancy-domination condition. The recurrence producing the endpoints need not implement Bellman iteration.

Conservative policy improvement. CPI mixes a current policy with a candidate to limit the effect of occupancy shift (Kakade & Langford, 2002; Kakade, 2003). Safe policy-iteration methods develop lower bounds and update rules under different approximation assumptions (Piotto et al., 2013; Metelli et al., 2021; Eshwar et al., 2025); TRPO uses a related surrogate with a trust-region

update (Schulman et al., 2015). We analyze one exact probability mixture and supply its evaluation-error term from the recurrent certificate. Keeping the signed defect and the exact selector-based occupancy bound distinguishes the roles of evaluator accuracy and policy shift. Sequence-level termination or approximate-optimality results require the additional oracle and coverage conditions of the corresponding CPI analyses. Table 5 compares these theorem scopes.

Structured decisions and verifier feedback. Learning-to-search methods cast structured prediction as sequential decision making (Daumé III et al., 2009; Ross et al., 2011; Chang et al., 2015); checker-based constraint tasks provide concrete edit environments (Mehta, 2021; Klein, 2022). Verifiable-reward reasoning systems such as DeepSeek-R1 use automatic correctness signals for RL (DeepSeek-AI, 2025). UPI-TRM studies such feedback in an edit MDP with an explicit finite-depth value evaluator. Its internal recurrence is distinct from unrolled planning over transitions (Tamar et al., 2016), trajectory search (Janner et al., 2021), and sequence-modeling approaches to control (Chen et al., 2021; Laskin et al., 2023): n counts computation within an evaluation, whereas K counts environment steps in a Bellman backup.

N.2 FINITE-ENDPOINT REMARKS

Remark N.1 (Noncontractive finite endpoints). Define

$$D_{n,m} := \|U_n - U_m\|_\infty. \quad (151)$$

Theorem 5.1 and the finite-path bound remain valid when a recurrent Lipschitz modulus is unavailable or when a valid modulus satisfies $L_z \geq 1$. Only the optional infinite-depth geometric specialization is then lost. A small $D_{n,m}$ alone does not certify value accuracy: the depth- m Bellman residual is still required. Nor does the theorem imply that error decreases monotonically with depth. For example, the scalar map $F(z) = -z$ is nonconvergent away from zero, while endpoints whose depths have the same parity agree exactly. This illustration is not a model of the learned system; it shows that the certificate concerns only the selected endpoint pair.

When the corresponding premises are all available, crude endpoint bounds may be combined as

$$D_{n,m} \leq \min \left\{ 2V_{\max}, 2L_V R, L_V C_z \sum_{j=n}^{m-1} L_z^j \right\}. \quad (152)$$

The first term assumes both endpoint outputs are bounded by $V_{\max}$; the second assumes both endpoint latents lie in B_R and the head is L_V -Lipschitz; the third assumes a uniform finite-path recurrent modulus $L_z \geq 0$ and first-increment bound C_z . If only some premises are proved, the minimum is taken only over the corresponding available terms. For $L_z \geq 1$, the last term need not shrink with n .

Remark N.2 (Frozen schedules and changing parameters). A fixed depth-indexed schedule $z^{(j+1)} = T_j(s, z^{(j)})$ may replace repeated use of one map in the abstract finite-endpoint and telescoping arguments, provided the schedule is frozen at the analyzed snapshot and produces bounded measurable U_n, U_m on the same Bellman domain. A recurrent mode or clock that changes across environment steps must be observed and included in the Markov state. If the recurrent maps, value head, policy, or Bellman operator change between defining the endpoints, collecting trajectories, evaluating the residual, and deploying the policy, the fixed-snapshot theorem does not apply as written. It is not a theorem about SGD, TD, BPTT, or adaptive parameter updates.

N.3 PROOF OF COROLLARY 5.3

Proof. Let $q_e(s, a) = \gamma \mathbb{E}[e(S') \mid s, a]$. Exact centering gives $b(s) = \mathbb{E}_{\pi_{\text{cand}}} q_e - \mathbb{E}_{\pi} q_e$. By Definition E.2, there is a common action set of full $\pi(\cdot \mid s)$ - and $\pi_{\text{cand}}(\cdot \mid s)$ -measure on which $q_e(s, a) \in [-\gamma\delta_V, \gamma\delta_V]$. The total-variation expectation inequality on this common essential range proves Eq. (12); taking a supremum or a d_π expectation gives the two corollaries. For a K -step estimator, $q_e^{(K)}(s, a) = \gamma^K \mathbb{E}[e(S_K) \mid s, a]$ has the corresponding essential range $[-\gamma^K\delta_V, \gamma^K\delta_V]$. $\square$

N.4 PROOF OF PROPOSITION 6.1

Proof. Add and subtract the target-network population backup:

$$U_q - \mathcal{T}_K^\pi U_q = [U_q - (r_K^\pi + \gamma^K P_\pi^K \bar{V})] + \gamma^K P_\pi^K (\bar{V} - U_q).$$

The triangle inequality gives the first bound in Eq. (19). Invariance makes P_π^K a Markov operator on $B_b(\mathcal{E})$, so it is nonexpansive in the pointwise sup norm and gives the second bound. $\square$

N.5 PROOF OF LEMMA 7.2

Proof. Realize each mixture decision with an independent Bernoulli selector. The event that decisions $0, \dots, t-1$ all select the current-policy component has probability $(1-\alpha)^t$ and, conditional on that event, produces the current-policy state law before decision t . Therefore

$$\text{TV}(\rho P_{\pi_\alpha}^t, \rho P_\pi^t) \leq 1 - (1-\alpha)^t.$$

Convexity of total variation and the geometric identity give

$$\begin{aligned} \text{TV}(d_{\pi_\alpha}, d_\pi) &\leq (1-\gamma) \sum_{t=0}^{\infty} \gamma^t [1 - (1-\alpha)^t] \\ &= \frac{\gamma\alpha}{1-\gamma+\gamma\alpha}. \end{aligned}$$

The exponent is t , not $t+1$, because occupancy is measured before decision t . The formula gives zero at $\alpha=0$ or $\gamma=0$ and γ at $\alpha=1$, and improves the generic bound because $1-\gamma+\gamma\alpha \geq 1-\gamma$. In the countable case this gives $2\gamma\alpha/(1-\gamma+\gamma\alpha) \leq 2\gamma\alpha/(1-\gamma)$. $\square$

N.6 RESIDUAL-CERTIFICATE COMPARISON TABLE

Table 7: Residual certificates inserted into the same exact-mixture CPI shell.

Certificate source	Value-error radius E	Evaluator term in the scalar CPI bound
Self-bootstrap at n	$\varepsilon_{\text{res},n}/(1-\gamma^K)$	$2\alpha\gamma\tau_\infty E/(1-\gamma)$
Target network at q	$R_{\text{targ},q}^P/(1-\gamma^K)$	same shell; the propagated lag remains inside $R_{\text{targ},q}^P$
Finite reference $n < m$	$\ U_n - U_m\ _\infty + R_{\text{targ},m}^P/(1-\gamma^K)$	same shell, evaluated with the depth- n advantage estimator

N.7 SPAN-BASED SAFE-STEP EXTENSION

Corollary N.3 (Span-based safe mixture step). *In the setting of Theorem 7.3, fix one measurable common policy-pair domain $\mathcal{C}_{\text{pair}}$ from Definition E.3, with $\mathcal{C}_\pi \subseteq \mathcal{C}_{\text{pair}}$. Use one fixed estimator $\hat{A}$ satisfying exact statewise centering on $\mathcal{C}_\pi$ for every $\alpha \in [0, 1]$, and define its fixed candidate-defect function*

$$b(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [\hat{A}(s, a) - A^\pi(s, a)]. \quad (153)$$

Neither the estimator nor this defect object depends on α . Define, once on $\mathcal{C}_{\text{pair}}$,

$$g(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [A^\pi(s, a)], \quad \Delta_g := \sup_{s \in \mathcal{C}_{\text{pair}}} g(s) - \inf_{s \in \mathcal{C}_{\text{pair}}} g(s), \quad (154)$$

and assume this function is bounded and measurable there. Let $\bar{\beta}_d$ be any finite, α -independent, established one-sided bound satisfying

$$\bar{\beta}_d \geq \mathbb{E}_{s \sim d_\pi} b(s). \quad (155)$$

Define

$$\hat{g} := \mathbb{E}_{s \sim d_\pi, a \sim \pi_{\text{cand}}(\cdot|s)} [\hat{A}(s, a)], \quad M_d := \hat{g} - \bar{\beta}_d. \quad (156)$$

Then, simultaneously for every $\alpha \in [0, 1]$,

$$\eta(\pi_\alpha) - \eta(\pi) \geq \frac{\alpha}{1-\gamma} \left[M_d - \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha} \right]. \quad (157)$$

Thus $\eta(\pi_\alpha) \geq \eta(\pi)$ is certified whenever

$$M_d \geq \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha}. \quad (158)$$

Three immediate choices preserve the same fixed defect object:

1. the exact signed defect, $\bar{\beta}_d = \mathbb{E}_{d_\pi} b(s)$;
2. the occupancy-averaged absolute defect, $\bar{\beta}_d = \mathbb{E}_{d_\pi} |b(s)|$;
3. the uniform specialization already used above, $\bar{\beta}_d = \varepsilon_{A,\text{cand}}$ from Eq. (30).

The third choice recovers the previous uniform statement as an immediate specialization. Appendix I.1 gives the exact positive-step threshold and all boundary cases. Under Corollaries 5.2 and 5.3, with the finite-reference estimator $\hat{A} = \hat{A}_{U_n}$, one may use $\mathbb{E}_{d_\pi} |b(s)| \leq 2\gamma B_{n,m} \mathbb{E}_{d_\pi} \tau(s)$ for the occupancy-averaged specialization, or $\varepsilon_{A,\text{cand}} \leq 2\gamma B_{n,m} \tau_\infty$ for the uniform specialization. The bound $\tau_\infty \leq 1$ recovers the previous uniform fallback.

N.8 EXTENDED DISCUSSION AND CERTIFICATION PROGRAM

Finite computation is the primary object. A recurrent evaluator can be analyzed at the depth where it is used. A second finite endpoint separates the effect of additional computation from agreement with the policy’s Bellman equation. This produces a value-error radius that feeds directly into advantage and CPI bounds. A small endpoint gap alone is insufficient: the reference must also have a small residual. Uniform contraction supplies a geometric specialization, but transient recurrent dynamics remain within the primary finite-reference analysis.

Persistent memory changes the decision process. The latent and remaining edit budget determine future decisions and termination. Including them in the state makes their role explicit. The shared frozen map is what allows the base, candidate, and their mixture to inhabit one augmented MDP. Increasing the comparison depth is then an evaluation operation, whereas increasing the deployed depth can change the transition kernel. A small Euclidean distance between two recurrent maps alone cannot control that change (Remark C.1).

What learned-model certification would require. A numerical certificate needs uniform residual control on the relevant successor domain, exact centering or a justified signed-defect bound, and control of any discrepancy between the deployed policy and the exact mixture. A target-network residual also needs its propagated lag. Finite-population maxima and regression losses are useful diagnostics, but they do not establish those uniform premises. Exact action enumeration resolves the action expectation, not state or latent coverage. The weighted-norm alternative makes that coverage requirement explicit.

Projection can bound the latent set without making these quantities easy to certify or small enough to be useful. It does not bound an arbitrary external state space, and products of global Lipschitz bounds or whole-ball interval bounds can be loose. A possible next step is to enumerate finite external states and actions, construct reachable latent enclosures, and preserve shared inputs and weights when bounding $U_n - U_m$. The same computation must also certify rewards, masks, successors, and Bellman expectations. Appendix L records the implementation details needed for that connection; such a learned-model verifier is not established here.

Scope and open questions. The results concern one frozen proposal. They do not establish convergence of its optimizer or a recursively promoted policy sequence, finite-sample coverage, near-optimality, or improvement on a learned task. The exact MDP calculation shows that the certificates can be informative and that retaining signed error can matter. Evaluating their diagnostic value for trained recurrent models, and obtaining practical uniform certificates, remain separate empirical and verification questions.